DEVELOPMENTAL BIOLOGY

An ERK-dependent molecular switch antagonizes fibrosis and promotes regeneration in spiny mice (*Acomys*)

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Although most mammals heal injured tissues and organs with scarring, spiny mice (*Acomys*) naturally regenerate skin and complex musculoskeletal tissues. Now, the core signaling pathways driving mammalian tissue regeneration are poorly characterized. Here, we show that, while immediate extracellular signal-regulated kinase (ERK) activation is a shared feature of scarring (*Mus*) and regenerating (*Acomys*) injuries, ERK activity is only sustained at high levels during complex tissue regeneration. Following ERK inhibition, ear punch regeneration in *Acomys* shifted toward fibrotic repair. Using single-cell RNA sequencing, we identified ERK-responsive cell types. Loss- and gain-of-function experiments prompted us to uncover fibroblast growth factor and ErbB signaling as upstream ERK regulators of regeneration. The ectopic activation of ERK in scar-prone injuries induced a pro-regenerative response, including cell proliferation, extracellular matrix remodeling, and hair follicle neogenesis. Our data detail an important distinction in ERK activity between regenerating and poorly regenerating adult mammals and open avenues to redirect fibrotic repair toward regenerative healing.

INTRODUCTION

Tissue damage or loss poses an immediate risk to organismal homeostasis. While the generation of scar tissue is a common response to limit the injury area in arthropods and vertebrates, some animals can completely regenerate lost tissue. A paradox is the seemingly uneven distribution of this regenerative response across major metazoan clades (*1*). Even in closely related species, tissue damage can elicit different healing outcomes. For instance, the ability to regenerate a tail varies widely among reptiles, while the mechanistic basis for this variation remains unknown (*2*). Among mammals, spiny mice (*Acomys* species) naturally regenerate complex tissues in the ear pinna, which includes full-thickness skin, hair follicles, adipose tissue, peripheral nerves, skeletal muscle, and cartilage. In contrast, laboratory mouse strains (*Mus*) and other murid rodents heal identical wounds with scar tissue characterized by excessive collagen deposition (*3*–*5*). Despite progress disentangling the mechanistic basis for these different healing responses, important questions remain. Tantamount among them is how the cellular response to injury is differentially regulated in regenerative versus fibrotic healing. Lineage tracing and transplantation experiments in murine skin wounds have suggested that specific fibroblast lineages or phenotypes are responsible for scar formation in laboratory mice (*6*–*8*), but the cellular basis for regeneration in spiny mice remains obscure.

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Vertebrate regeneration occurs as a complex series of overlapping processes where inflammation and wound closure are followed by inflammatory resolution and sustained cell cycle progression of aggregated progenitor cells (a transition referred to as blastema formation) (*9*). Progenitor cells and their progeny ultimately undergo differentiation and morphogenesis to functionally replace the damaged or missing tissue. The degree to which healing initiators, activated in response to injury, directly influence blastema formation, persistent cell cycle progression, and morphogenesis remains unknown. Because enhanced regenerative ability has likely evolved de novo in mammals (*10*), comparing the healing response of identical injuries in regenerating and nonregenerative species offers an opportunity to identify cell-autonomous and cell-extrinsic drivers of mammalian regeneration. Given the highly conserved role that extracellular signal-regulated kinase (ERK)-mediated signaling plays during the metazoan injury response (*11*–*13*), it is an attractive target to explore the molecular basis for the existence of different responses to similar injuries.

In the context of tissue healing, ERK activation is known to regulate the re-epithelialization of skin wounds (wound closure) (*14*), and recent work has demonstrated that mitogen-activated protein kinase (MAPK)/ERK–mediated wound signals jumpstart the regenerative response in planarian flatworms (*15*). Tissue regeneration occurs at a relatively slow rate in mammals and salamanders (compared to invertebrates and zebrafish), providing a unique opportunity to dissect how temporal ERK activation regulates the cellular response to injury throughout wound healing, progenitor cell proliferation, and tissue reconstitution. By analyzing ERK activation and its role during fibrotic repair in *Mus* and regenerative healing in *Acomys*, we show that (i) immediate ERK activation is a conserved feature of the injury response independent of the healing outcome; (ii) ERK activation is sustained at a significantly high level after injury only in *Acomys* (compared to *Mus*); (iii) during regeneration, ERK is activated in keratinocytes and a diverse array of stromal cells; (iv) inhibiting ERK activation in *Acomys* after re-epithelialization shifts the normal regeneration program toward fibrotic healing; and (v) ectopic ERK activation confers regenerative traits to normally nonregenerative *Mus* tissues. Together, our data demonstrate that high ERK activity acts as a molecular switch between regeneration and scarring and partially explains why some closely related mammalian species may exert very different cellular responses to the same injury.

RESULTS

Injury-induced ERK activation is a common feature of wound healing in nonregenerative (*Mus*) and regenerative (*Acomys*) mammalian tissue injuries

To test our hypothesis that ERK activity might differentially regulate healing outcomes in *Acomys* and *Mus*, we first explored wound-induced ERK activation [phosphorylated ERK (pERK)] using a 4-mm ear punch injury (*4*, *5*) and a pERK-specific antibody (Fig. 1A and fig. S1, A and B). In relation to the long axis of the ear pinna, the ear punch generated a proximal and a distal segment where healing occurs (Fig. 1A). While active ERK was nearly absent in uninjured tissue from both species (the exception being a few hair follicle bulb cells), 10 min after injury, we observed high pERK levels spreading away from the injury site, mainly in the epidermis in *Mus* and *Acomys* (Fig. 1, B to D). Relative to *Mus*, ERK was more broadly activated in *Acomys* from 1 hour post-injury (hpi) (Fig. 1, E and F) with significantly higher levels in the ventral epidermis, perichondrium, and cartilage (Fig. 1G). A similar phenomenon occurred in the distal ear segment during the early phase of the injury response (fig. S1, C to F). Endothelial and mural cells, nerves, chondrocytes and perichondral cells, keratinocytes, and mesenchymal and muscle cells all immediately activated ERK at the wound site in the proximal ear segment (Fig. 1H and fig. S2, A to E). Together, these data show that pERK acts as an immediate sensor in response to tissue damage in regeneration-incompetent (*Mus*) and regeneration-competent mammals (*Acomys*).

Sustained high ERK activity is a hallmark of tissue regeneration

Following the initial burst of ERK activity in response to injury, we evaluated ERK activation after the initial wound healing phase [at 10 days after injury (D10)] when the injury response transitions to fibrosis in *Mus* or regeneration in *Acomys* (*4*). In *Mus*, ERK activity significantly decreased from D10 [Fig. 1, I (left) and J (top), and fig. S3, A to F], and we observed very few scattered pERK$^+$ cells throughout the fibrotic tissue in either the proximal [Fig. 1, I (left) and J (top), and fig. S3, A to E] or distal ear segments (fig. S3, G to I). The intensity of pERK signal was also much lower compared to *Acomys* (fig. S3F). In contrast to *Mus*, we found persistent and robust ERK activity in the proximal ear segment during regeneration in *Acomys* [Fig. 1, I (left) and J (top), and fig. S3, A to F]. However, in the *Acomys* distal ear segment, we observed almost no ERK activity from D5 to D25 (fig. S3, G to I). This was consistent with previous observations of asymmetric regeneration in *Acomys*, where new tissue formation occurs preferentially in the proximal ear segment, whereas during fibrotic repair in *Mus*, scar tissue is produced equally around the circular punch (*4*, *5*).

As new tissue began to fill the open hole in *Acomys*, we observed activated ERK throughout the epidermal and mesenchymal connective tissue (blastema) (Fig. 1I, left, and fig. S3, C to E). At D20, approximately 70% of the pERK signal was localized in nerves, labeled by βIII-tubulin, at the blastema tip (fig. S3J, left). Notably, ERK was also activated in cycling (proliferating cell nuclear antigen, PCNA$^+$) mesenchymal cells and basal keratinocytes at the blastema tip (fig. S3J, right). Together, these data demonstrate that after an initial burst of injury-induced ERK activation, robust pERK activity is maintained after D10 to a much higher degree in *Acomys* and mostly in the proximal ear segment coincident with persistent cell cycle progression and new tissue formation.

To test whether higher and sustained ERK activation is a more general feature of regenerating tissues, we also analyzed ERK activation in *Mus* and *Acomys* injured back skin and infarcted hearts. Spiny mice regenerate full-thickness skin excisions and exhibit enhanced cardiac repair in response to myocardial infarction (*3*, *16*–*18*). We found that after an initial burst of injury-induced ERK activation in *Mus* and *Acomys* dorsal skin (fig. S4), pERK levels were significantly higher in the epidermis of *Acomys* compared to *Mus* at D35 after full-thickness skin wounds [Fig. 1, I (middle) and J (middle), and fig. S4]. We observed a similar temporal pattern in the epicardium and subjacent myocardium of infarcted hearts with a significantly higher number of pERK$^+$ nuclei in *Acomys* compared to *Mus* at the injury site 100 days after myocardial infarction [Fig. 1, I (right) and J (bottom), and fig. S5, A and B], indicating that sustained high ERK activity is a key feature of a pro-regenerative healing response in *Acomys*.

ERK activation is required independently for re-epithelialization and regeneration

To investigate the potential role of ERK activation for wound healing and regeneration, we treated spiny mice with the highly selective small-molecule inhibitor mirdametinib [PD0325901, currently used in clinical trials (*19*), hereon referred to as PD] that prevents ERK phosphorylation and, thereby, its activation (*20*). The phosphorylation of both ERK1/2 isoforms (P-p44/P-p42) was strongly reduced upon PD treatment in the epidermis and dermis at 6 hpi (Fig. 2, A and B). A series of inhibition experiments was designed to inhibit the early and late phase of ERK activation separately or simultaneously (Fig. 2C). In prior studies (*4*, *5*), re-epithelialization was documented to occur by D12, and new tissue growth closed the ear hole by D40 (Fig. 2, D and E). In contrast, continuous ERK inhibition blocked regeneration and we observed incomplete re-epithelialization and a persistent scab until at least D20 (Fig. 2, D and E). Similarly, animals treated with the early PD scheme exhibited severely delayed wound closure and impaired regeneration (Fig. 2, D and F).

Because wound closure is necessary to resolve wound healing and promote blastema formation (*21*, *22*), we specifically targeted the late phase of ERK activation associated with new tissue formation by starting the PD treatment at D12 after re-epithelialization occurred (Fig. 2C). Spiny mice undergoing late PD treatment completed re-epithelialization normally, but regeneration was still markedly compromised compared to the control group, supporting a specific role for ERK activity during regeneration (Fig. 2, D and E). These data demonstrate that injury-induced ERK activation is necessary for wound closure, and its sustained high activity is critical for regeneration.

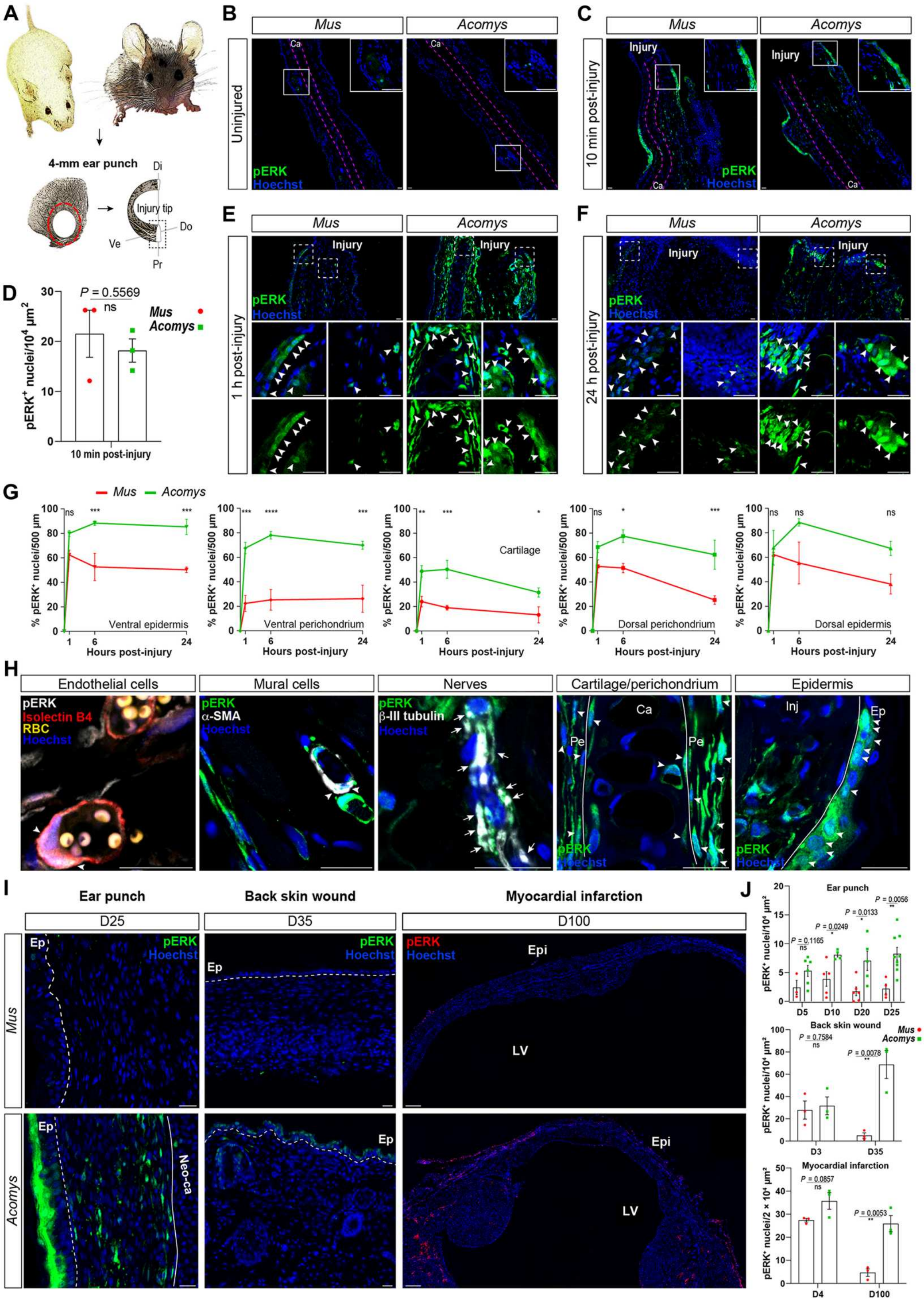

Fig. 1. ERK is immediately activated upon injury in regenerators and poor regenerators but only sustained during regeneration. (**A**) Comparison of *Mus* (left) and *Acomys* (right) ear punch (red circle): proximal (Pr), distal (Di), ventral (Ve), and dorsal (Do). (**B** to **F**) Representative images of injury-induced ERK activation: undamaged tissue (insets, hair follicles) (B), 10 min (C) and relative quantification (D), 1 hpi (E) and 24 hpi (F) in the proximal segment; insets, bottom. (**G**) Quantification of pERK$^+$ nuclei over total nuclei, 500 μm to the injury tip (*P* values left to right: 0.0863, 0.0004, 0.0005; 0.0002, <0.0001, 0.0003; 0.0043, 0.0005, 0.0376; 0.2248, 0.0174, 0.0009; 0.9858, 0.0609, 0.1166). (**H**) Injury-induced cell type–specific ERK activation in *Acomys* (within 6 hpi). (**I**) Representative images of ERK activation at late time points after punch (left), back skin injury (middle), and myocardial infarction (right). pERK$^+$ keratinocytes in the epidermal basal and upper layers, papillary dermal fibroblasts, mesenchymal cells, and cells in the neo-cartilage (Neo-ca) of *Acomys* blastema (left). pERK$^+$ cells in the *Acomys* epidermis and newly formed hair follicles (middle), and in the epicardium and subjacent myocardium in the infarcted area (right). Arrowheads, pERK$^+$ nuclei; arrows, cytoplasmic pERK. Injury tip (Inj), cartilage (Ca), perichondrium (Pe), epidermis (Ep), myocardial infarction (MI), epicardium (Epi), left ventricle (LV), autofluorescent red blood cells (RBC), nuclei (Hoechst). (**J**) Quantification of pERK$^+$ nuclei. Scale bars, 20 μm, except 200 μm (I, right). Each dot represents one biological replicate. Data are represented as means ± SEM; ns, not significant, *$P < 0.05$, **$P < 0.01$, ***$P < 0.001$, ****$P < 0.0001$.

ERK inhibition shifts the *Acomys* tissue injury response toward fibrotic repair

To characterize the ERK-dependent molecular events driving blastema formation and regeneration in *Acomys*, we selectively inhibited ERK activation after re-epithelialization (D12) and performed RNA sequencing (RNA-seq) analyses on the proximal ear segments collected at D15 (Fig. 3A and fig. S6A). The efficiency of the PD treatment was confirmed immunohistochemically at D15 using the total pERK signal to quantify ERK inhibition (fig. S6, B and C). To evaluate whether systemic drug administration might have nonlocal effects, we quantified dying cells in healing tissue and in tissue distant from the injury. While we observed significantly more dying cells locally at the injury site after ERK inhibition, we found almost no dying cells distant to the injury supporting a local effect of

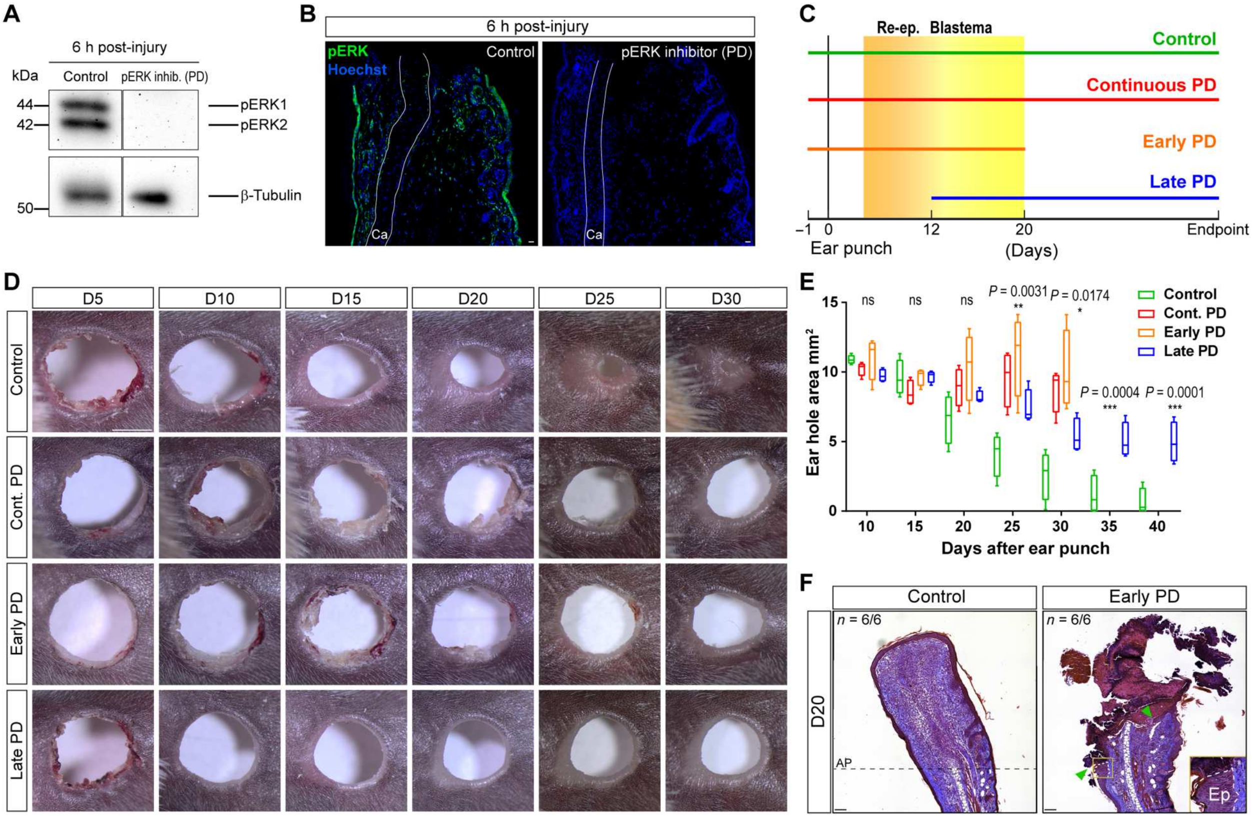

Fig. 2. Wound closure and regeneration are independently impaired upon ERK inhibition in *Acomys*. (**A** and **B**) Pharmacological inhibition of ERK activation by mirdametinib (PD) treatment, confirmed by Western blot (P-p44/P-p42, phosphorylated ERK1/2 isoforms; β-tubulin as loading control, $n = 3$ per group) (A) and immunofluorescence (pERK, green, $n = 3$ per group) at 6 hpi; Ca, cartilage (B). (**C**) Experimental design of ERK inhibition in *Acomys*. PD treatment started after re-epithelialization (Re-ep) at D12 in the Late PD group. (**D**) Representative images of ear hole closure for the same *Acomys* ear after 4-mm punch. Proximal to distal ear segment, left to right, respectively. (**E**) Box-and-whisker plots of ear hole closure: bottom whisker corresponds to minimum, box bottom to 25th percentile, box middle to median, box top to 75th percentile, top whisker to maximum ($n = 4$ per group; area of the two ears averaged per mouse). Two-way repeated-measures (RM) ANOVA with Sidak's multiple comparisons test. (**F**) Representative images of Masson's trichrome staining showing severe impairment of wound closure in the Early PD group. New tissue growth and de novo appendage formation in control ears (left, AP amputation plane), open wound with scab formation in Early PD group (arrowheads indicate the two wound margins, right). Scale bars, 20 μm (B), 2 mm (D), and 100 μm (F). ns, not significant, $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$.

the drug treatment (fig. S6, D to H). RNA-seq analysis comparing PD-treated tissues to control ear tissues at D15 revealed 265 differentially expressed genes (DEGs) [false discovery rate (FDR) < 0.1], 120 of which could be categorized into processes associated with blastema formation and expansion (Fig. 3B) (*21*, *23*). These processes included gene sets associated with cell proliferation, immature wound epidermis (specialized epithelium of activated keratinocytes), production of a pro-regenerative extracellular matrix (ECM), resolution of excessive inflammation, and nerve growth (*21*).

Consistent with previous work demonstrating a role for ERK signaling in cell cycle progression through the G_1-S checkpoint in vitro (*11*), RNA profiling revealed that mRNA levels of proteins involved in the G_1-S transition, including *cyclin D1*, were down-regulated after PD treatment (Fig. 3, B and C). In addition, mRNA levels of the G_2-M–specific *Cyclin B2* and the mitotic regulator *Nusap1* were down-regulated upon ERK inhibition, while suppressors of cell cycle entry, such as cyclin-dependent kinase inhibitor 1B (*Cdkn1b*, $p27^{Kip1}$), known to promote G_1 arrest, were up-regulated (Fig. 3, B and C). Gene set enrichment analysis (GSEA) identified the E2F target collection as the top down-regulated gene set upon ERK inhibition compared to the control group at D15 (Fig. 3D). E2F factors are transcriptional regulators of cell cycle genes (*24*). To further examine the potential for ERK-mediated cell cycle progression during regeneration, we quantified the number of cells actively undergoing S phase (5-ethynyl-2′-deoxyuridine, EdU^+) and those undergoing mitosis (phospho-histone H3, $pHH3^+$). Cellular analysis revealed a significantly strong reduction in EdU^+ and $pHH3^+$ cells upon ERK inhibition at D15 (Fig. 3, E to G). Together, these data demonstrate that ERK activation is required for sustained cell proliferation in vivo during regeneration and suggest that ERK drives active cell cycle progression by modulating the expression of *cyclin D1*, *cyclin B2*, and E2F downstream targets.

In addition to promoting cell proliferation during regeneration, we tested whether other key hallmarks of blastema formation were impaired upon ERK inhibition (Fig. 3B). One of these hallmarks is the formation and maintenance of an immature, transitional epidermis (wound epidermis), a critical step for the expansion of new tissue (*4*, *25*). The wound epidermis of axolotls and spiny mice retains an embryonic-like expression pattern of keratins, specifically the expression of wound-induced keratin 17 (K17) and K5, a marker for dividing basal keratinocytes (*3*, *4*, *26*). Conversely in

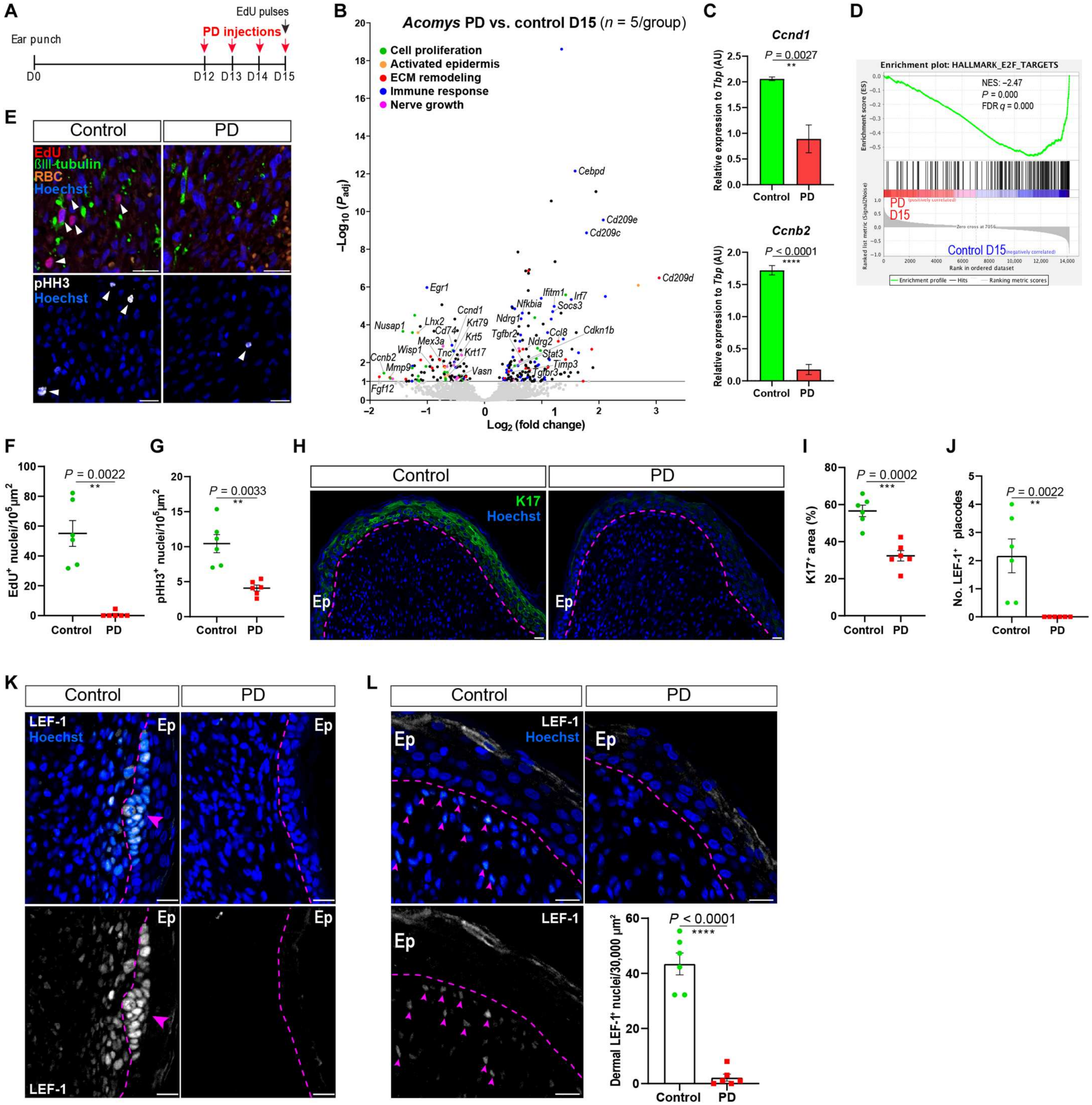

Fig. 3. ERK activation is required to orchestrate a regenerative response in *Acomys* upon injury. (**A**) Study design for ERK inhibition using PD. (**B**) Volcano plot displaying DEGs from PD-treated versus control injured tissue samples with color-coded categories indicated at top left. (**C**) Reverse transcription quantitative PCR (RT-qPCR) validation ($n = 5$ per group) for *Ccnd1* (cyclin D1) and *Ccnb2* (cyclin B2). (**D**) GSEA plot showing a drastic down-regulation of E2F targets in the PD group compared to control at D15. Normalized enrichment score (NES), FDR, nominal *P* value ($n = 5$ per group). (**E** to **G**) Representative images and quantification of EdU$^+$ (red, top) and pHH3$^+$ (white, bottom) nuclei in the injury area (D15) ($n = 6$ per group); βIII-tubulin (green) (top). (**H** and **I**) Representative images and quantification of K17$^+$ area (green) as percent of positive epidermis in the injury tip at D15 ($n = 6$ per group). (**J**) Quantification of epidermal LEF-1$^+$ placodes at D15 ($n = 6$ per group). (**K**) Representative images of LEF-1$^+$ nuclei (white) and hair placode (arrowheads) at D15. (**L**) Representative images and quantification of dermal LEF-1$^+$ nuclei (white, arrowheads) in the injury tip (D15). Pink dashed line, epidermis (Ep)–dermis boundaries. Autofluorescent red blood cells (RBC), arbitrary units (AU), nuclei (Hoechst, blue). Scale bars, 20 μm. Data are represented as means ± SEM; $^{**}P < 0.01$, $^{***}P < 0.001$, $^{****}P < 0.0001$.

Mus, after a transient, wound-induced increase in K17 expression, K17 declines to basal levels as the tissue undergoes fibrotic repair (*4*). Consistent with the RNA-seq analysis showing that *Krt17* and *Krt5* expression was down-regulated in PD-treated samples (Fig. 3B), K17 protein levels were markedly decreased in keratinocytes overlying the wound upon ERK inhibition in *Acomys* at D15, and after that, re-epithelialization occurred in both groups (Fig. 3, H and I). These data demonstrate that sustained ERK activity is necessary to maintain high K17 levels in the wound epidermis concomitant with a transitional keratinocyte profile.

During complex tissue regeneration in *Acomys*, hair follicle neogenesis occurs from the amputation plane toward the ear punch center as new tissue production closes the open hole (*3*). K79 is expressed by migratory keratinocytes involved in hair follicle morphogenesis (*27*), and RNA profiling showed that *Krt79* was down-regulated following ERK inhibition (Fig. 3B). To assess de novo hair follicle formation, we evaluated the presence of the hair placode marker lymphoid enhancer binding factor 1 (LEF-1) and found no LEF-1$^+$ nuclei in the epidermis or LEF-1$^+$ epidermal hair placodes in PD-treated *Acomys* compared to the control group (Fig. 3, J and K). LEF-1 belongs to the T cell factor (TCF)/LEF family of transcription factors that mediate the nuclear response of the Wnt/β-catenin pathway (*28*). We also tested whether ERK inhibition affected LEF-1 levels in the dermis where the induction of *Lef1* expression in dermal fibroblasts has been shown to stimulate enhanced hair follicle neogenesis in normally scar-producing wounds (*29*, *30*). We observed dermal cells positive for nuclear LEF-1 beneath the hair germ in the papillary dermis (Fig. 3K) and in the blastema tip of the control group during regeneration (Fig. 3L). In PD-treated animals, LEF-1$^+$ nuclei were almost entirely absent from the dermis at D15 (Fig. 3, K and L). These data show that LEF-1$^+$ mesenchymal cells naturally emerge during regeneration in the *Acomys* ear pinna and that hair follicle neogenesis strongly depends on ERK activity.

Next, to gain insights into the ERK-dependent matrisome, we analyzed the DEGs between PD and control groups against the murine ECM atlas (http://matrisomeproject.mit.edu) (Fig. 4A) (*31*). Among the identified ERK-dependent matrisome was the matrix metalloproteinase-9 (MMP-9), an ECM-degrading enzyme that can degrade collagen fibers and promote cell migration (*32*, *33*). *Mmp9* is also highly expressed during newt limb and spiny mouse regeneration (*4*, *34*). In addition to *Mmp9*, we identified other matrisome genes whose expression was altered upon ERK inhibition. Among those were genes encoding secreted factors, including fibroblast growth factor 12 (*Fgf12*) and C-C motif chemokine ligand 8 (*Ccl8*), ECM regulators, such as tissue inhibitor of metalloproteinase 3 (*Timp3*), core matrix proteins, including Wnt-1 inducible signaling protein 1 (*Wisp1*) and tenascin C (*Tnc*), and a C-type lectin receptor (*Cd209d*) (Fig. 4, A and B). *Timp3*, a tissue inhibitor of metalloproteinases that binds and irreversibly inactivates collagenases and metalloproteinases, such as MMP-9, was up-regulated in the PD group (Figs. 3B and 4A). Notably, *Fgf12* and pro-regenerative genes, including *Tnc*, the Wnt/β-catenin downstream effector *Wisp1*, and *Mmp9*, were significantly down-regulated upon ERK inhibition (Fig. 4, A and B). Wisp1 acts as a secreted matricellular signal regulating ECM production and regeneration (*35*). In stark contrast, *Cd209d* and profibrotic genes, including *Ccl8* (*36*), were up-regulated in the PD group (Fig. 4, A and B). Recent work identified a pro-regenerative ECM profile in *Acomys*, with enrichment for *TnC*, *Mmp9*, and low levels of collagen type I alpha 1 chain (*Col1a1*) (*4*). In contrast, *Mus* fibrotic repair was characterized by abundant *Col1a1* and reduced *Tnc* and *Mmp9* expression. While we found MMP-9$^+$ cells in the blastema and among basal keratinocytes of the wound epidermis during regeneration, the number of MMP-9$^+$ cells significantly decreased upon ERK inhibition (Fig. 4, C and D).

Because of the collagen-degrading activity of MMP-9 and the close association of type 1 collagen with fibrosis (*37*), we quantified collagen deposition in the proximal and distal ear segments in PD-treated spiny mice compared to the control group at D20 and D45. We found that type 1 collagen protein levels were significantly higher following ERK inhibition in the proximal ear segment compared to the control group at D20, and this difference persisted at D45 (Fig. 4, E and F). Notably, no differences were observed between the distal ear segments of the PD and control groups at D20 (Fig. 4, E and F). Among the enriched subsets of the Gene Ontology Cellular Components (GO-CC), the gene sets related to ECM and collagen were found to be significantly enriched upon PD treatment (fig. S7). Furthermore, transforming growth factor–β (TGF-β) signaling is a master regulator of ECM assembly and fibrogenesis and is up-regulated in organ fibrosis (*38*). Accordingly, PD-treated spiny mice exhibited significantly higher levels of TGF-β receptor 2 (*Tgfbr2*) and 3 (*Tgfbr3*) expression and reduced levels of vasoneurin (*Vasn*), a negative regulator of TGF-β signaling (Fig. 4G) (*39*). Our ECM profiling data, together with the reduced proliferation, indicate that loss of ERK activity shifts the *Acomys* healing trajectory toward fibrotic repair.

Last, the outcome of the injury response toward fibrotic or regenerative healing relies on the timely resolution of inflammation. An exacerbated or dysregulated inflammatory response can lead to chronic fibrosis and scarring (*40*). Upon ERK inhibition from D12, *Acomys* ear tissue increased levels of key inflammatory genes at D15, including CCAAT enhancer–binding protein delta (*Cebpd*), signal transducer and activator of transcription 3 (*Stat3*), suppressor of cytokine signaling 3 (*Socs3*), nuclear factor of kappa light polypeptide gene enhancer in B cells inhibitor alpha (*Nfkbia*), interferon regulatory factor 7 (*Irf7*), and interferon-induced transmembrane protein 1 (*Ifitm1*) (fig. S8A). Accordingly, ERK inhibition from D12 resulted in a significantly higher number of infiltrating leukocytes, positive for CD45 (fig. S8, B and C). GSEA revealed that gene sets associated with inflammation, including tumor necrosis factor–α (TNF-α)/nuclear factor κB (NF-κB) signaling, were up-regulated at 24 hpi (early injury response characterized by acute inflammation) (*40*) but down-regulated in control *Acomys* punched ears at D15 during regeneration (fig. S8D). In contrast, the inflammatory response and, particularly, TNF-α/NF-κB and interleukin-6 (IL-6)/Janus kinase (JAK)/signal transducer and activator of transcription 3 (STAT3) signaling were up-regulated in the punched ear tissue upon 3-day ERK inhibition (from D12 to D15) compared to the regenerating control at D15 (fig. S8E). This suggests a reengagement of inflammatory pathways upon ERK inhibition, as the tissue exhibited hallmarks of fibrosis. Considered together, our data show that when ERK is inhibited after re-epithelialization is complete in *Acomys* tissue, the normal regenerative response is skewed toward a scar-prone wound resolution, typical of *Mus* fibrotic healing.

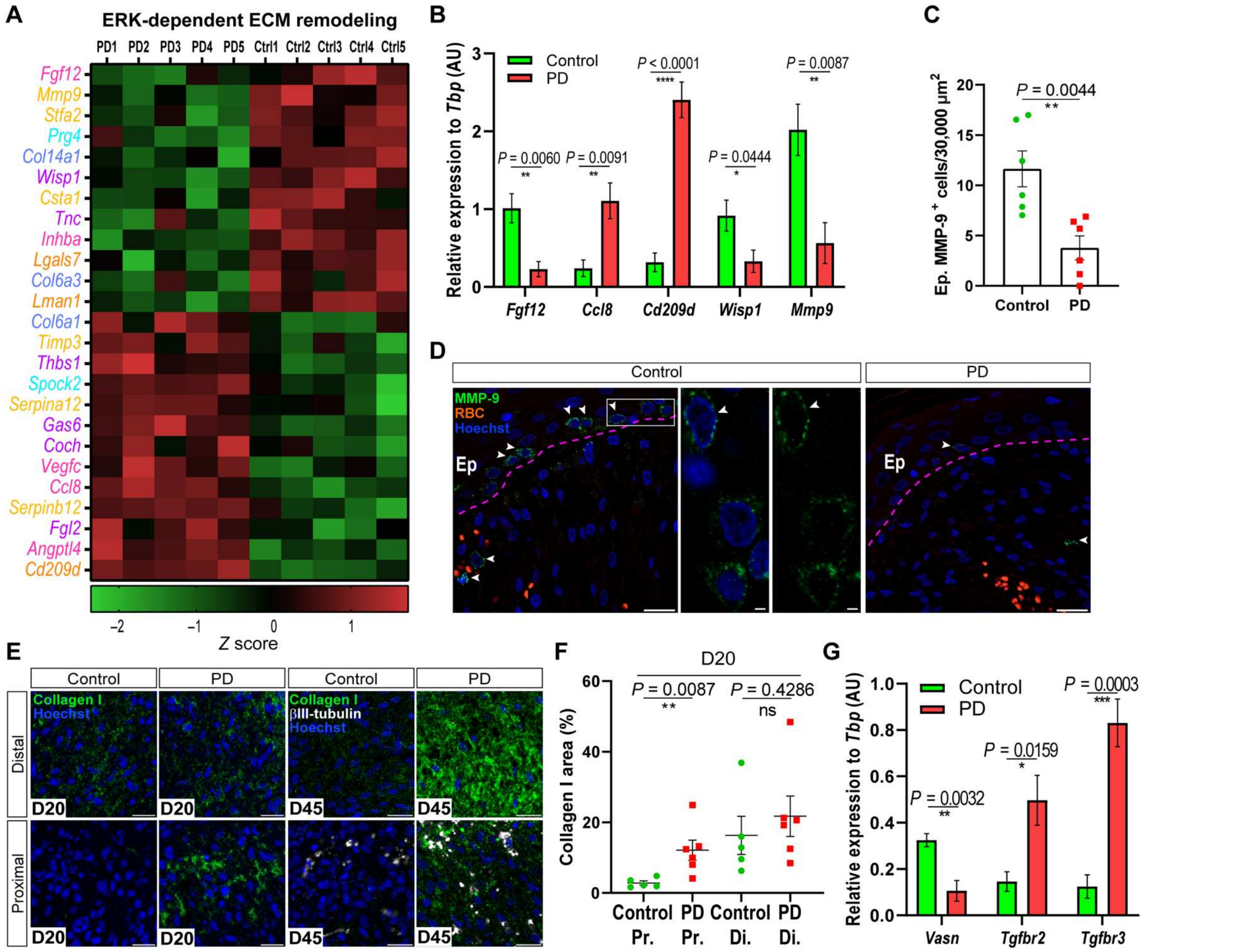

Fig. 4. ERK inhibition converts normally regeneration-associated matrisome into a profibrotic ECM profile. (**A**) ERK-dependent ECM remodeling. Matrisome-associated divisions were subcategorized into secreted factors (pink), ECM-affiliated proteins (dark orange), and ECM regulators (light orange). The core matrisome included proteoglycans (cyan), ECM glycoproteins (purple), and collagens (cobalt blue). (**B**) RT-qPCR validation of select matrisome genes (D15, $n = 5$ per group). (**C** and **D**) Representative images and quantification of MMP-9$^+$ cells (green) in the injury tip at D15 ($n = 6$ per group). Zoom-in of MMP-9$^+$ cells in the basal layer of the epidermis (Ep). (**E** and **F**) Representative images and quantification of collagen 1a1 (green) in the injury area of the proximal and distal ear segments at D20 ($n = 5$ control, 6 PD) (left) and D45 ($n = 4$ per group) (right); βIII-tubulin (white). (**G**) RT-qPCR of TGF-β signaling components (D15, $n = 5$ per group). Scale bars, 20 μm; 2 μm in inset (D). Gene expression value normalized by z score transformation ($n = 5$ per group) (A). Data are represented as means ± SEM; ns, not significant, $*P < 0.05$, $**P < 0.01$, $***P < 0.001$, $****P < 0.0001$.

The injury-induced ERK targets EGR-1 and Fos are specifically sustained during blastema formation and regeneration

To investigate whether ERK activation coordinates a regenerative program that might explain its diverse effect on blastema formation in terms of cell proliferation, keratinocyte polarization, ECM remodeling, and resolution of inflammation, we next analyzed transcription factor binding sites enriched in ERK-dependent genes. When we applied GSEA, we recovered significant changes in the TGF-β signaling pathway and in the activator protein-1 (AP-1) transcription factor network, including *Mmp9*, *Ccnd1*, and early growth response 1 (*Egr1*), which were significantly down-regulated upon ERK inhibition at D15 (Fig. 5A and fig. S9A). When we interrogated our RNA-seq dataset against collections of transcription factor targets (*41*), the consensus AP-1 binding site, TGANTCA, was among the most enriched motifs together with the binding site for the nuclear factor of activated T cells (NFAT), a binding partner for AP-1 proteins (*42*), and the serum response factor (SRF), which can activate the transcription of AP-1 factors (fig. S9B) (*43*).

AP-1 factors induce cell cycle progression in vitro by modulating cyclins and E2F, transcriptional regulators of cell cycle genes (*24*). Consistent with these findings, we found that *Cyclin D1*, *Cyclin B2*, and E2F target genes were down-regulated upon ERK inhibition in vivo (Fig. 3, B to D). The AP-1 motif is enriched in a group of injury-responsive enhancers driving both injury and regenerative responses in fish (*44*). Moreover, EGR-1 regulates chromatin accessibility and is required for regeneration in acoel flatworms (*45*). EGR-1 and Fos are immediate-early genes (IEGs) rapidly induced upon stress (*46*), and we found that these proteins were activated in the nuclei of keratinocytes and mesenchymal cells at the injury site 6 hours after ear punch (Fig. 5B and fig. S9, C and D). Nuclear EGR-1$^+$ and Fos$^+$ cells (a subset double positive for pERK) populated the

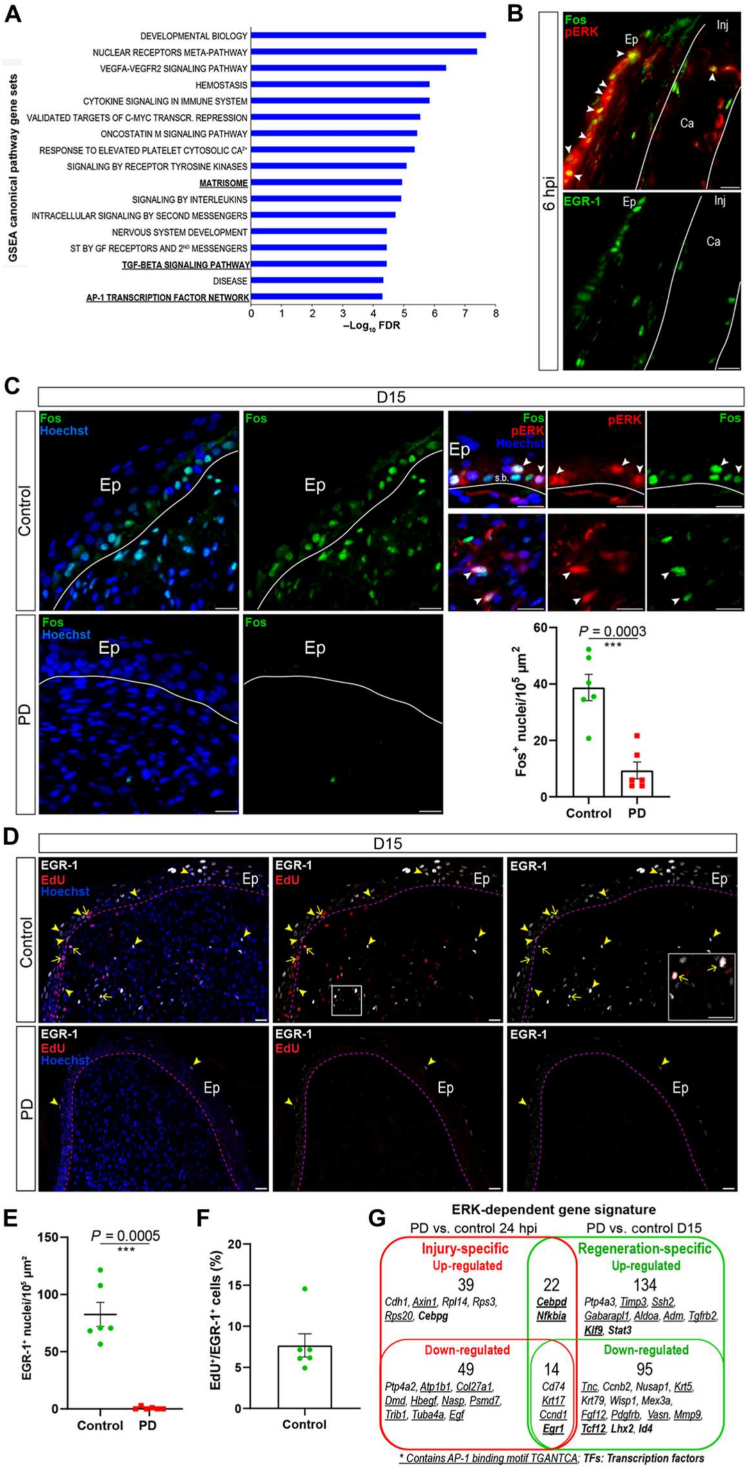

Fig. 5. ERK activity induces Fos and EGR-1 downstream targets during spiny mouse regeneration. (**A**) GSEA canonical pathway analysis showing statistically significant gene sets (*y* axis) enriched upon ERK inhibition. *X* axis indicates $-\log_{10}$ FDR statistical significance (n = 5 per group). (**B**) Representative images of IEGs Fos (top) and EGR-1 (bottom) induced immediately after punch at the injury tip (arrowheads, double-positive cells). (**C**) Representative images of IEG Fos activated during regeneration (top) and inhibited in PD (bottom). Costaining of Fos and pERK, top right (arrowheads, double-positive cells in the epidermis (Ep), top, and dermis, bottom). Quantification of Fos^+ nuclei (bottom right). (**D**) Representative images of the IEG ERK downstream target EGR-1 (white) and EdU (red). Arrowheads indicate $EGR\text{-}1^+$ cells, and arrows indicate $EdU^+/EGR\text{-}1^+$ cells. Inset shows $EdU^+/EGR\text{-}1^+$ double-positive cells at the blastema tip in the control group. (**E**) Quantification of $EGR\text{-}1^+$ nuclei at D15 (n = 6 per group). (**F**) Quantification of proliferating $EGR\text{-}1^+$ cells over total $EGR\text{-}1^+$ cells at D15, after 9-hour EdU exposure (n = 6). (**G**) Venn diagram showing genes up- and down-regulated in PD at 24 hpi and D15. In the intersection, genes induced by injury and sustained during regeneration (bottom). Bold, transcription factors; underscored, genes containing the AP-1 binding motif. White line, cartilage (B); white and pink dashed line, epidermis (Ep)–dermis boundaries (C and D, respectively); signal transduction (ST), growth factor (GF), injury (Inj); cartilage (Ca); stratum basale (s.b.), nuclei (Hoechst, blue). Scale bars, 20 μm. Data are represented as means ± SEM; $^{***}P < 0.001$.

Acomys blastema during regeneration, and their number was markedly reduced upon ERK inhibition at D15 (Fig. 5, C to E). Moreover, approximately 8% of all $EGR\text{-}1^+$ cells were actively cycling (colabeled with EdU), detected after three pulses of EdU injection over 9-hour exposure (Fig. 5F). These data indicate that the ERK targets EGR-1 and Fos are not only expressed immediately after injury but also activated by sustained high ERK activity during regeneration.

To identify an injury-specific and a regeneration-specific ERK gene signature, we compared the top DEGs between control and PD-treated *Acomys* ear tissue isolated at 24 hours after injury (injury-specific) with those isolated at D15 (regeneration-specific) (Fig. 5G). Thirty-five of 124 top DEGs activated in the early injury response (24 hours after injury) and 77 of 265 top DEGs activated at D15 contained at least one putative AP-1 binding site in their promoter region (Fig. 5G and table S1). Among the set of genes down-regulated upon ERK inhibition at 24 hpi and D15, we found that *Keratin17*, *Ccnd1*, and *Egr1* expression was induced in the early phase after injury and maintained during regeneration at D15, showing the same sustained pattern of ERK activation (Fig. 5G and table S1). Together, these data support pERK acting as one of the key linkages connecting early injury signals to the later stages of regeneration via a mechanism where sustained high ERK activity maintains the expression of some injury-induced genes and also induces regeneration-specific gene expression. Moreover, our data support that the transition between the early injury response and regeneration could be partially mediated by AP-1 binding activity.

ERK activation is regulated by FGF and ErbB-2 signaling and correlates with regeneration rate

Ear punches in *Acomys* regenerate asymmetrically with new tissue production occurring from the proximal area of the hole (*21*). In line with this, we found that ear holes closed faster when deploying the 4-mm ear punch nearer the head compared to a more distally placed punch in the other ear of the same spiny mouse (Fig. 6, A and B). While proximal holes closed within 20 days (95.58% ± 3.13 of hole area), only one-third of the area was filled by new tissue in the distal holes at D20 (26.75% ± 3.74 of hole area) (Fig. 6, A and B). The regeneration rate of the proximal ear punch was 215 ± 34 μm/day between D10 and D15, almost three times faster than a distal ear punch (70 ± 14 μm/day) (fig. S10A). As ERK phosphorylation is required for *Acomys* tissue regeneration (Fig. 2), we asked whether the faster regeneration rate in proximal holes correlated with higher ERK activation. We found significantly higher levels of ERK activation in proximal versus distal injuries, which positively correlated with ear closure rate (Fig. 6C and fig. S10, B and C). These data support that the extent of ERK activation is a key determinant for tissue growth and regeneration and suggest that exogenous factors present in the injury microenvironment might control the dynamics of ERK signaling.

Since nerves are necessary to maintain cell cycle progression and are crucial sources of the mitogens fibroblast growth factor 2 (FGF-2) and neuregulin 1 (NRG-1), two molecules that rescue axolotl limb regeneration in the absence of nerves (*47*), we sought to investigate whether these ligands were expressed in the *Acomys* blastema. We deployed ligand analysis using the top DEGs between control and PD group at D15 against the Library of Integrated Network–based cellular signatures (LINCS L1000 Ligand perturbation) (*48*). This analysis supported a role for FGF and NRG-1 and revealed that the epidermal growth factor (EGF) and FGF family of ligands (*49*) were the most enriched in the control group at D15 (fig. S10D, left). Conversely, proinflammatory cytokines were the top hits in the PD group, providing further evidence of the reengagement of inflammation upon 3-day ERK inhibition (fig. S10D, right). Screening for these ligands, we found that NRG-1$^+$ cells populated the dermis and the epidermis (Fig. 6D, left, and fig. S11A), including muscle fibers near the cartilage (fig. S11B) and the papillary dermis of the growing blastema below the epidermis (fig. S11C). Similarly, FGF-2$^+$ cells were scattered in the papillary dermis and blastema tip (Fig. 6D, right, and fig. S11, D to F) and were highly abundant in hair follicles and at the border between the preexisting tissue and the newly formed blastema (fig. S11, E to G).

To functionally explore whether these signaling molecules regulated ERK signaling during normal regeneration, we pharmacologically inhibited FGFR and ErbB-2 receptors by using AZD4547 (*50*) and mubritinib (*51*), respectively, beginning at D12 (after re-epithelization) and continuing through D20. Quantifying ear hole closure, we found that ear punch regeneration was strongly impaired compared to the control group [one-way analysis of variance (ANOVA), $F = 5{,}179$, $P = 0.0078$] (Fig. 6, E and F), and notably, ERK activation significantly decreased by 66 and 73% in the FGFR and ErbB-2 inhibitor groups, respectively (one-way ANOVA, $F = 8.816$, $P = 0.0006$) (Fig. 6, G and H, and fig. S12A). Cell proliferation and the ERK targets EGR-1 and Fos were also markedly reduced compared to the regenerating control group, indicating that ErbB-2, FGFR, and ERK signaling converge on similar downstream targets (Fig. 6, I to L, and fig. S12, B to D).

Given these results, we asked whether the administration of FGF-2 and NRG-1 would stimulate ERK activation in normally nonregenerating *Mus* ear pinna injuries. Following 4-mm ear punch, we locally implanted microbeads soaked with recombinant FGF-2 and NRG-1 after re-epithelialization at D10 and found that the combination of these two ligands led to a significant increase in tissue growth compared to the control group at D20 (fig. S13, A and B, and Fig. 7, A and B). To determine whether FGF-2/NRG-1 administration enhanced ERK activation, we quantified total ERK activation, the number of nuclear pERK$^+$ cells, and EdU incorporation at D20. In contrast to the control phosphate-buffered saline (PBS)–soaked beads, the punched ears implanted with FGF-2/NRG-1 beads showed significantly enhanced ERK activation and an increased number of EdU$^+$ cells in the epidermis and dermis (Fig. 7, C and D, and fig. S13, C to F). This manifested into a higher cell density around the FGF-2/NRG-1–soaked beads (Fig. 7E). We also observed hair follicle neogenesis at the injury tip in *Mus* wounds at D20, demonstrating that the increased earhole closure occurred through regenerative healing and not simply through increased scar tissue production (Fig. 7E). Together, these data indicate that FGFR and ErbB-2 act as upstream ERK regulators during composite tissue regeneration in spiny mice and that the combination of FGF-2 and NRG-1 can induce ERK activation in *Mus* injuries.

Single-cell transcriptomics identifies common and distinct pro-regenerative ERK signatures in specific cell types

On the basis of our immunohistochemical data demonstrating ERK activation in a diverse array of cell types, we next used single-cell transcriptomics [single-cell RNA sequencing (scRNA-seq)] to identify downstream ERK targets in a cell type–specific manner to ask how ERK activity more precisely regulates regeneration. To accomplish this, we excised 4-mm ear punches and inhibited separately ERK, FGFR, and ErbB-2 signaling in *Acomys* from D12 (after re-epithelialization) to D15, when new tissue starts to regenerate. Using this experimental design, we performed scRNA-seq on 43,700 cells isolated from treated and control *Acomys* injuries, which allowed us to identify 16 cell types in uniform manifold approximation and projection (UMAP) space (Fig. 8A and fig. S14, A and B). We first applied GSEA among the three inhibitors deployed in *Acomys* to identify core ERK-activated gene sets expressed across cell types. We found that FGFR, ErbB-2, and ERK signaling converged on 22 hallmark gene sets, including KRAS signaling, E2F and Myc targets, TGF-β and TNF-α signaling, and G_2-M

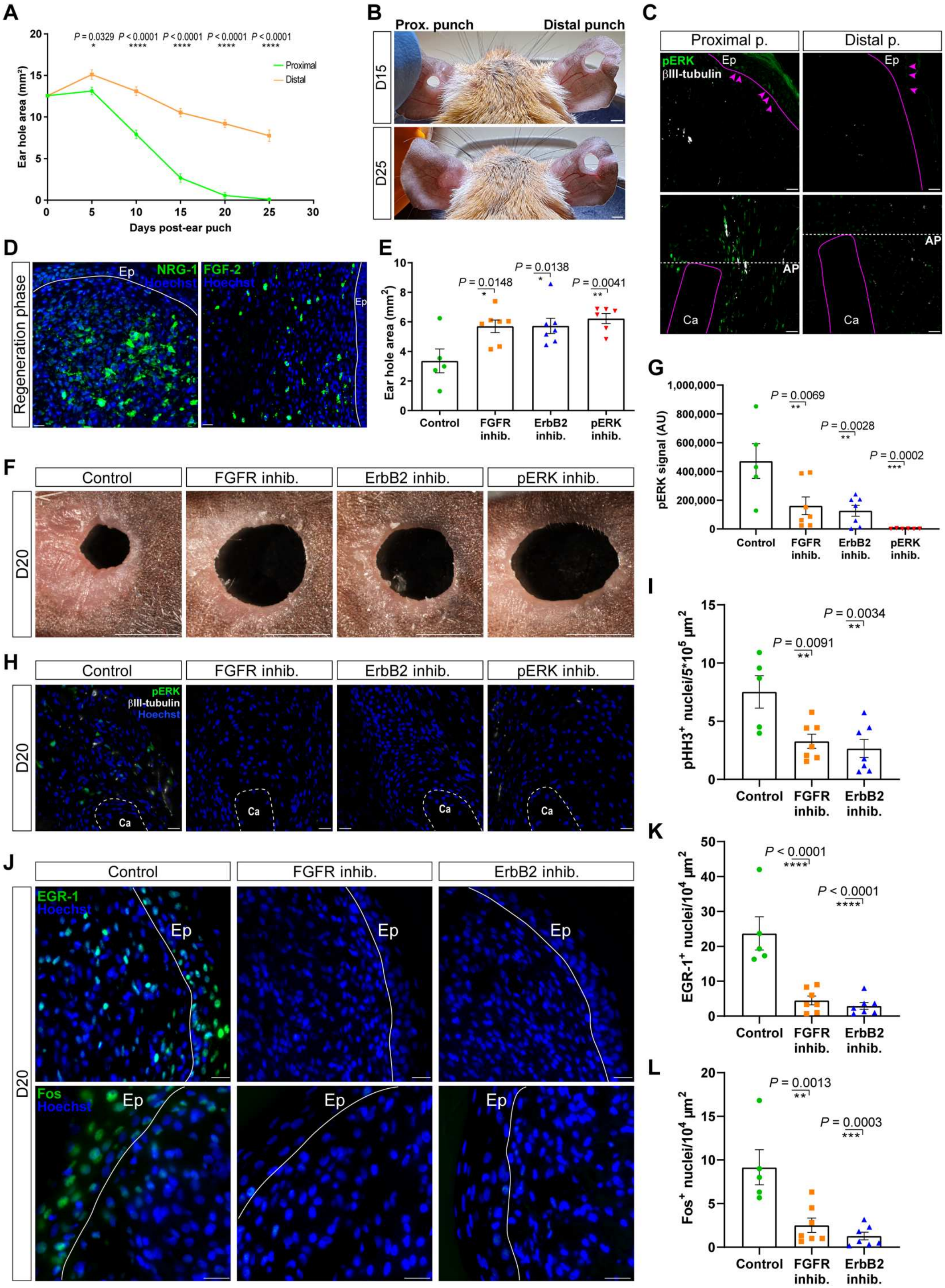

Fig. 6. ERK activity correlates with regeneration rate and responds to FGFR and ErbB-2 receptor activity. (**A**) Quantification of the hole area in proximal and distal 4-mm ear punches ($n = 6$). (**B**) Representative images of hole closure after proximal (left) and distal (right) 4-mm punch. (**C**) Representative images of pERK (green, arrowheads) at injury tip (top) and amputation plane (AP) (bottom), βIII-tubulin (white); Ep, epidermis. (**D**) Representative images of NRG-1 (left) and FGF-2 (right) ligands expressed in the blastema, including the papillary dermis, during regeneration. (**E** and **F**) Quantification and representative images of ear hole closure after FGFR ($n = 7$), ErbB-2 ($n = 7$), or ERK inhibition ($n = 6$) compared to the control group ($n = 5$) at D20. (**G** and **H**) Quantification and representative images of ERK activation (pERK, green) in control ($n = 5$), FGFR ($n = 7$), ErbB-2 ($n = 7$), or pERK ($n = 6$) inhibitor groups; arbitrary unit (AU), βIII-tubulin (white), intact cartilage (Ca) of the pre-amputation plane. (**I**) Mitotic cell ($pHH3^+$) quantification upon FGFR and ErbB-2 receptor inhibition. (**J** to **L**) Representative images of the wound edge and quantification of the IEG ERK targets EGR-1 (top) and Fos (bottom) upon inhibition; white line, epidermal-dermal boundary. Each dot represents one biological replicate. Scale bar, 20 μm, except for 2 mm (B and F). Two-way RM ANOVA with Sidak's multiple comparisons test (A), one-way ANOVA with Dunnett's multiple comparisons test (E, G, I, K, and L). Data are represented as means ± SEM; *$P < 0.05$, **$P < 0.01$, ***$P < 0.001$, ****$P < 0.0001$.

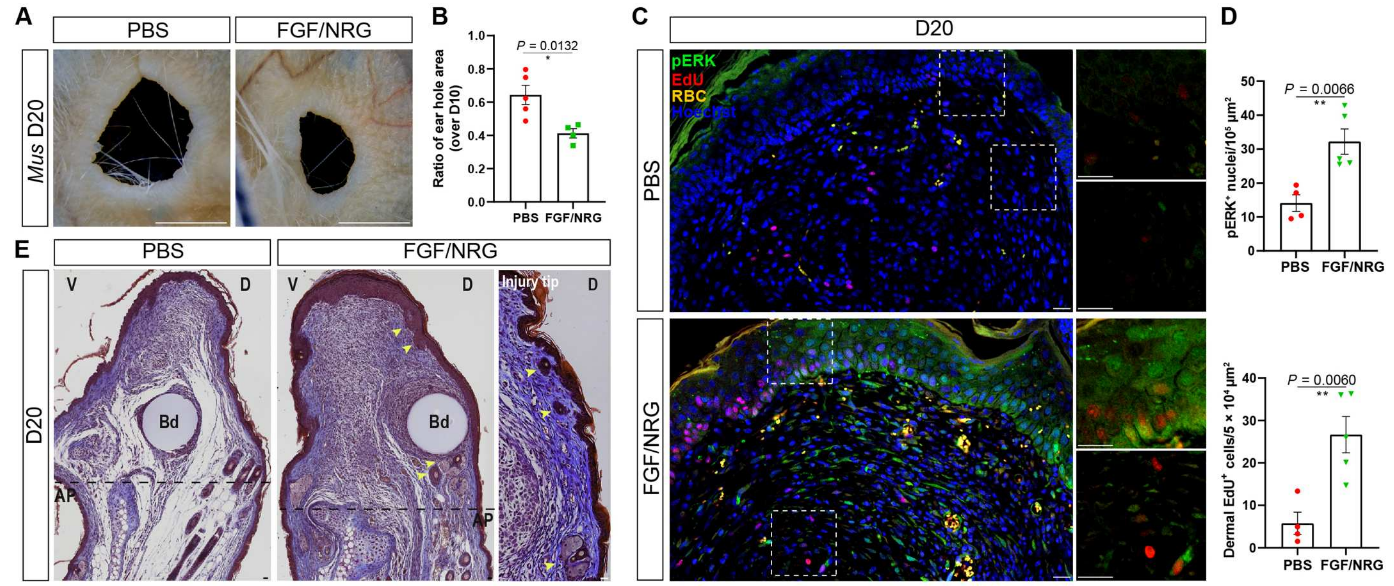

Fig. 7. ERK activators FGF-2 and NRG-1 stimulate a regenerative response in *Mus*. (**A** and **B**) Representative images and quantification of ear hole area after 4-mm ear punch following implantation of microbeads soaked with FGF-2 and NRG-1. PBS-soaked beads were used as controls. The beads were implanted at D10 after re-epithelialization occurred. (**C** and **D**) Representative images and quantification of ERK activation (pERK$^+$ nuclei) and cell proliferation (EdU$^+$) in the FGF-2/NRG-1–treated samples compared to control at D20. (**E**) Representative Masson's trichrome staining of the bead-implanted ears and newly formed appendages at the injury tip (top) in FGF-2/NRG-1–treated wounds (right, arrowheads indicate de novo hair follicle formation); bead (Bd), ventral (V), dorsal (D) (n = 4 PBS, n = 5 FGF-2/NRG-1). Amputation plane (AP), autofluorescent red blood cells (RBC, orange), nuclei (Hoechst, blue). Scale bar, 20 μm, except 2 mm (A). Each dot represents one biological replicate. Data are represented as means ± SEM; *$P < 0.05$, **$P < 0.01$.

checkpoint, 21 canonical pathways, including matrisome and keratinization, 35 transcription factor targets, including AP-1, Myc, and TEAD2 (*52*), and 71 genes differentially expressed during regeneration (Fig. 8, B and C, and fig. S14, C and D). In addition, markers of G_1-S and G_2-M phases of the cell cycle were markedly reduced in response to FGFR, ErbB-2, and ERK inhibition (fig. S14E).

Next, we used an unbiased approach to identify specific cell populations in which ERK was activated during regeneration. We leveraged four independent databases, three of which contained effectors of the known ERK targets Myc, E2F, and AP-1 (*41*, *53*–*55*) and a phospho-ERK target database (*56*). To identify Myc, E2F, and AP-1 targets in our datasets, we first interrogated the genes that were significantly up-regulated during regeneration and down-regulated upon ERK, FGFR, and ErbB-2 inhibition against the Myc, E2F, and AP-1 databases (*41*, *53*, *55*). Consistently, we found that ERK, Myc, AP-1, and E2F targets were down-regulated upon ERK, FGFR, and ErbB-2 inhibition (Fig. 8D and fig. S14F). These data supported genes and gene sets previously identified as ERK targets (*56*) and reinforced genes we identified from our bulk RNA-seq experiments as involved in regeneration (Figs. 3, B to D, and 5A and fig. S9B).

Examining Myc, AP-1, and E2F target genes, we found that they had a similar overlap and were highly expressed in basal keratinocytes, stromal cells, and immune cell types (fig. S14F). Specifically, we interrogated the ERK database (*56*) to identify a common set of ERK targets that are positively regulated in response to injury during regeneration and down-regulated upon FGFR, ErbB-2, and ERK inhibition (Fig. 8E). We used this consensus target list as a proxy for ERK activation in individual cell types ("ERK target module") (Fig. 8F). First, the cycling (IFE C) and basal (IFE B) interfollicular epidermis, which are composed of actively proliferating keratinocytes and their stem cell pools, respectively, showed the highest level of consensus ERK activation, followed by cells forming hair follicles (Fig. 8F). Subsets of each of the stromal cell types (fibroblasts, chondrocytes, dermal papilla, and endothelial cells) also exhibited high ERK activity. Among immune cell populations, subclusters of mainly T cells, B cells, and resident macrophages activated ERK during regeneration (Fig. 8F and fig. S14F). We also analyzed the consensus ERK target list in cells exposed to the inhibitors and found that many genes were down-regulated in different cell types, an observation that reinforced these genes as ERK targets during regeneration (Fig. 8E and fig. S14G). Next, we assessed FGF and EGF receptor expression across cell types and found that most receptors were broadly expressed in at least 50% of keratinocytes and stromal cells with ErbB-2 also expressed in most immune cell types (fig. S14H). Last, we specifically looked at stromal cells and keratinocytes, cell types that are central players in blastema formation and new tissue growth. Within these populations, we identified shared [i.e., junction plakoglobin (*Jup*), lamin A/C (*Lmna*), and catenin β1 (*Ctnnb1*)] and cell type–specific [i.e., adenosine triphosphatase (ATPase) family AAA domain containing 2 (*Atad2*), immediate early response 3 (*Ier3*), and enolase 1 (*Eno1*)] ERK targets down-regulated upon ERK, FGFR, and ErbB-2 inhibition (fig. S15, A and B, and table S2). These data support ERK activity as a key driver of regeneration and that it acts through common and cell type–specific targets in keratinocytes and stromal cells.

To further elucidate whether an ERK-dependent regenerative signature could be activated in *Mus* injuries, we implanted FGF-2/NRG-1–soaked beads into punched *Mus* ears at D10 to stimulate

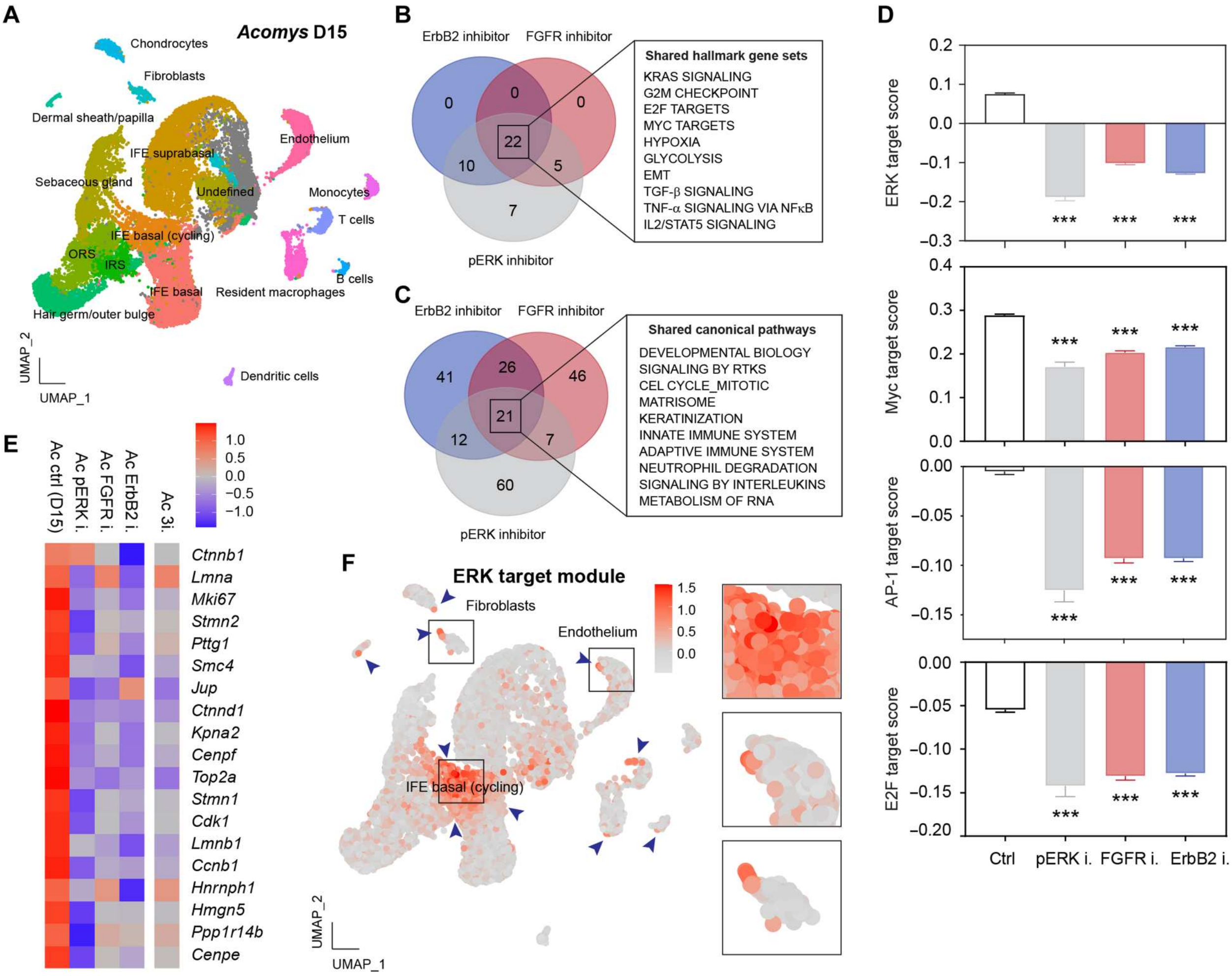

Fig. 8. Discrete cell subsets exhibit ERK activation during regeneration in *Acomys*. (**A**) Transcriptional profile–based cell clustering of all *Acomys* tissue samples harvested at D15 representing control, ERK, FGFR, and ErbB-2 inhibition. (**B** and **C**) Venn diagram showing shared hallmark gene sets (B) and pathways (C) affected following FGFR, ErbB-2, and ERK inhibition. FGFR, ErbB-2, and ERK signaling converge on similar gene sets (intersection); receptor tyrosine kinase (RTK). (**D**) Down-regulation of ERK targets, Myc targets, AP-1 targets, and E2F targets (top to bottom) upon FGFR, ErbB-2, and ERK inhibition. (**E**) Heatmap for consensus ERK targets up-regulated during regeneration and inhibited in response to ERK, FGFR, and ErbB-2 inhibition. Gene expression shown individually for each inhibitor and all three combined inhibitors (Ac 3i.) to exclude off-target effects from any single inhibition modality. This gene set represents the ERK target module used in (F). (**F**) Cell type–specific expression of ERK target module during regeneration; arrowheads indicate cell subsets exhibiting ERK activation, which are labeled in (A). ***$P < 0.001$.

ERK signaling as described above and harvested ear tissues at D15 (Fig. 7). We performed scRNA-seq on 12,746 cells recovered from the *Mus* bead implants (PBS control and FGF-2/NRG-1 groups) and identified the same 16 cell types found in *Acomys* (fig. S16, A to E). After batch-correcting the data across both species for all control and treatment conditions (see Materials and Methods), we analyzed DEGs recovered across the regenerative conditions (i.e., normal *Acomys* regeneration and FGF-2/NRG-1 *Mus* bead implantation) compared to the fibrotic conditions (i.e., three different inhibitors in *Acomys* and normal *Mus* injuries undergoing fibrotic repair). Analyzing keratinocytes (fig. S17) and stromal cell subsets (fig. S18), including fibroblasts (Fig. 9, A to C) from both species, we found consensus gene sets and genes that were up-regulated during regeneration in *Acomys* and also during ERK stimulation in *Mus* while down-regulated upon ERK, FGFR, and ErbB-2 inhibition in *Acomys* and *Mus* normal fibrotic repair (Fig. 9, A to C, and figs. S17, A to E, and S18, A to E). Using GSEA, we specifically found that among the genes up-regulated in keratinocytes across the three pairs of comparison, many of the consensus genes proved to be either Myc or AP-1 targets (fig. S17, C to E). We observed a similar phenomenon with most consensus genes among endothelial cells, chondrocytes, dermal papilla, and fibroblasts represented by Myc targets (Fig. 9, B and C, and fig. S18, C to E).

Fibroblasts are among the most important cells directing the healing outcome toward a regenerative or scarring response (*6*–*8*). EnrichR (*48*) analysis of our bulk RNA-seq data at D15 showed that myofibroblasts were the dominant cell type affected by ERK inhibition, suggesting a phenotypic shift away from the pro-regenerative *Acomys* fibroblasts (fig. S19A). To better understand whether ERK activity correlates with a regenerative phenotype, we further analyzed ERK activation in fibroblasts from our dual species scRNA-seq data. First, we identified two topologically distinct fibroblast

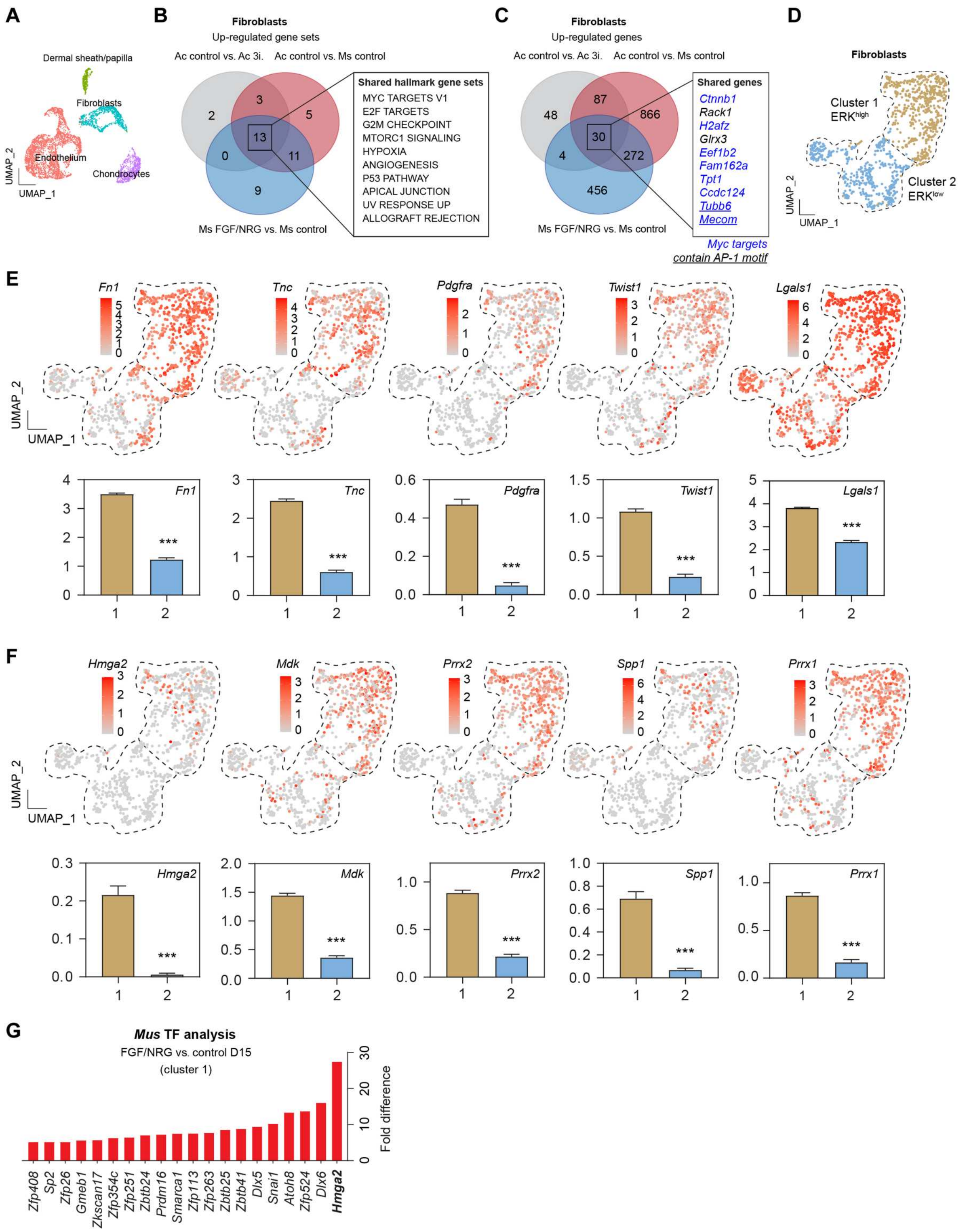

Fig. 9. Single-cell transcriptomics identifies an ERK-dependent transcriptional program in fibroblasts. (**A**) UMAP projection of stromal cell populations from batch-corrected data representing all D15 tissue samples from *Acomys* and *Mus* (control D15 *Acomys* and *Mus*; ERK, FGFR, and ErbB-2 inhibition in *Acomys*; FGF-2/NRG-1 stimulation in *Mus*). (**B** and **C**) Venn diagrams displaying in the intersection the gene sets (B) and DEGs (C) up-regulated in the regenerative scenarios during *Acomys* regeneration and upon FGF-2/NRG-1 stimulation in *Mus* injuries while down-regulated following ERK, FGFR, and ErbB-2 inhibition and *Mus* fibrotic repair in fibroblasts. (**D**) Fibroblast subclustering. Cluster 1 is enriched in genes up-regulated upon ERK activation and down-regulated upon ERK inhibition, while cluster 2 is enriched in genes up-regulated upon ERK inhibition and repressed upon ERK activation (see fig. S19B for details). The gene lists enriched in the ERK^{high}-activating fibroblasts and ERK^{low} subsets are visualized on the combined fibroblast UMAP. (**E** and **F**) Expression of select pro-regenerative and developmental genes in ERK^{high} fibroblast subset (cluster 1) represented in subsequent panels. (**G**) Transcription factor enrichment in fibroblasts upon FGF-2/NRG-1 stimulation compared to PBS control in *Mus* injuries at D15. ***P < 0.001.

subclusters through unsupervised analysis (Fig. 9D). Second, we mapped our previously defined ERK-activating module (corresponding to genes up-regulated upon ERK activation and down-regulated upon ERK inhibition) and found that it was enriched in cluster 1 compared to cluster 2 (fig. S19B, left). Next, we mapped our ERK-repressing module (defined as genes inhibited upon ERK activation) and found that it was enriched in cluster 2 compared to cluster 1 (fig. S19B, right). Genes associated with fibrosis, including fatty acid–binding protein 5 (*Fabp5*) (*57*), integral membrane protein 2B (*Itm2b*) (*58*), and *Serpinb5* (*59*), were up-regulated in the ERK^{low} cluster 2 (fig. S19C). Conversely, genes expressed during regeneration across animal models and tissue types, including those encoding fibronectin (*4*, *60*) and Tnc (*4*), the mesenchymal progenitor marker platelet-derived growth factor receptor α (PDGFR-α) (*61*), and the ERK target Twist1, which is also highly expressed during embryonic skin development (*62*), were all enriched in the ERK^{high} fibroblast cluster 1 and lowly expressed in cluster 2 (Fig. 9, E and F). In addition to known pro-regenerative genes, we also identified genes enriched in cluster 1 that have been associated with self-renewal, embryonic dermis, and tissue remodeling, including high-mobility group AT-hook 2 (*Hmga2*) (*63*), secreted phosphoprotein 1 (*Spp1*) (*64*), midkine (*Mdk*) (*65*, *66*), lectin galactoside–binding soluble 1 (*Lgals1*) (*67*), paired related homeobox 1 (*Prrx1*) (*23*, *68*), and its paralog *Prrx2* (Fig. 9, E and F). The chromatin modifier HMGA2 mediates transcription activation and is highly expressed during embryonic development in undifferentiated tissues, while it is detectable only in the stem cell pool in adult tissues (*63*, *69*). *Hmga2* was highly expressed in the ERK-activating fibroblast cluster 1 and showed an approximately 30-fold increase in the *Mus* wounds when treated with the ERK activators FGF-2/NRG-1 compared to the PBS control *Mus* at D15 (Fig. 9G). The FGF-2/NRG-1 mix increased the expression of blastema-associated transcription factors, including the ERK target Twist1 mediator *Snai1* (Fig. 9G) (*62*). Together, these data provide further evidence that the newly formed blastema is composed of distinct ERK-responsive cell subsets, mainly in the basal layer of the epidermis, hair follicles, and fibroblasts. In particular, an ERK-activating fibroblast subcluster expressed pro-regenerative markers, suggesting that FGF-2/NRG-1 can stimulate an ERK-directed regeneration program.

ERK activators stimulate a regenerative response in scarring wounds

To test whether the FGF-2/NRG-1–soaked beads could induce a pro-regenerative microenvironment similar to what we observed during normal regeneration in spiny mice, we harvested ear pinna tissue after 4-mm punch and assessed K17 levels in the epidermis comparing treated and control *Mus* injuries. Consistent with increased ear hole closure in *Mus* ears after bead implantation (Fig. 7, A and B), FGF-2/NRG-1–soaked beads increased K17 protein levels in the wound epidermis at D20 (Fig. 10, A and B), resembling the sustained K17 activation observed during *Acomys* regeneration (Fig. 3H, left) (*4*, *60*). K17 levels showed a strong positive correlation with keratinocyte proliferation and ERK activation (fig. S20, A and B), and ERK activation positively correlated with cell proliferation in the dermis of the injury tip (fig. S20C). Notably, we detected augmented fibronectin 1 (FN1) levels in the newly produced mesenchyme compared to the PBS control group at D20 (Fig. 10, C and D), and the FGF-2/NRG-1 mix stimulated a significant increase in MMP-9$^+$ cells at the injury tip (Fig. 10, E and F). Thus, the FGF-2/NRG-1 mix triggered ERK activation and cell proliferation among stromal cells, promoted an FN1- and MMP-9–enriched microenvironment, and led to increased cell proliferation and K17 activation in the *Mus* wound epidermis.

Next, we asked whether the transplantation of FGF-2/NRG-1–soaked microbeads would affect the cellular composition of the injury microenvironment. We analyzed the abundance of alpha smooth muscle actin (α-SMA)$^+$ myofibroblasts, cells that are strongly associated with fibrosis (*70*), and of PDGFR-α$^+$ mesenchymal cells, which are essential for mammalian digit tip regeneration and transition to blastemal cells after injury (*61*). The injury tip of the control *Mus* showed high levels of α-SMA and low levels of PDGFR-α at D20. By contrast, the injury microenvironment of the FGF-2/NRG-1–treated *Mus* was characterized by a reduction of α-SMA$^+$ cells and an increase of PDGFR-α$^+$ cells, a subset of which was positive for pERK (Fig. 10, G to J, and fig. S20, D and E). These data, together with the formation of new hair follicles (Fig. 7E), demonstrate that ERK activity is capable of shifting a fibrotic repair program to exhibit hallmarks of regeneration in *Mus* wounds. Additional evidence to support this conclusion was provided by our analysis of differentially expressed ERK upstream regulators and downstream effectors during *Acomys* regeneration and *Mus* fibrotic repair (*4*), suggesting that ERK signaling acts as a major hub directing different healing outcomes (fig. S20F). In this regard, FGF-2, NRG-1, and their respective receptors showed a strong up-regulation in *Acomys* during regeneration at transcriptional level, while their expression was limited in the *Mus* counterpart (fig. S20F). Furthermore, the expression of the dual specificity phosphatases *Dusp4*, *Dusp6*, and *Dusp8* that are all endogenous suppressors of ERK activation was up-regulated in *Mus* during fibrotic repair and down-regulated in *Acomys* during regeneration (fig. S20F). These data lend support to the findings that the limited ligand availability and low ERK activity in *Mus* injuries contributes to a failed regenerative response.

Last, to provide additional evidence that ERK activation could stimulate a regenerative response, we took advantage of an adeno-associated virus (AAV)–mediated gene transfer system to transduce the *Mus* ear pinna with an ERK activator. The AAV8 serotype efficiently transduces mesenchymal skin progenitors (*71*, *72*), and we made use of an AAV8 vector carrying the MAPK/ERK kinase $1^{S218D/S222D}$ (MEK1DD) transgene encoding a constitutively active MEK1 mutant, which, when deployed, leads to ERK activation (*73*). Following re-epithelialization in response to the 4-mm ear punch injury, we injected AAV8-MEK1DD particles locally near the ear hole at D10 using two doses: a low dose [approximately 10^9 genome copies (gc)] and a high dose (5×10^{10} to 5×10^{11} gc) in two separate groups. Control ears received AAV8–green fluorescent protein (GFP) particles. After confirming the efficiency of the procedure and AAV transduction (Fig. 10K), we asked whether the AAV8-MEK1DD gene transfer could induce a regenerative response similar to our FGF-2/NRG-1 bead transplantation experiments. We found that the delivery of MEK1DD via AAV transduction enhanced ear closure at D20 in a dose-dependent manner (Fig. 10L and fig. S20G). The levels of pERK were sustained over time with this method, and de novo hair follicle neogenesis was observed in the newly formed tissue [Fig. 10, M and N (top), and fig. S20, H and I]. Similar to the increased fibronectin levels observed upon FGF-2/NRG-1 stimulation (Fig. 10D), AAV8-MEK1DD

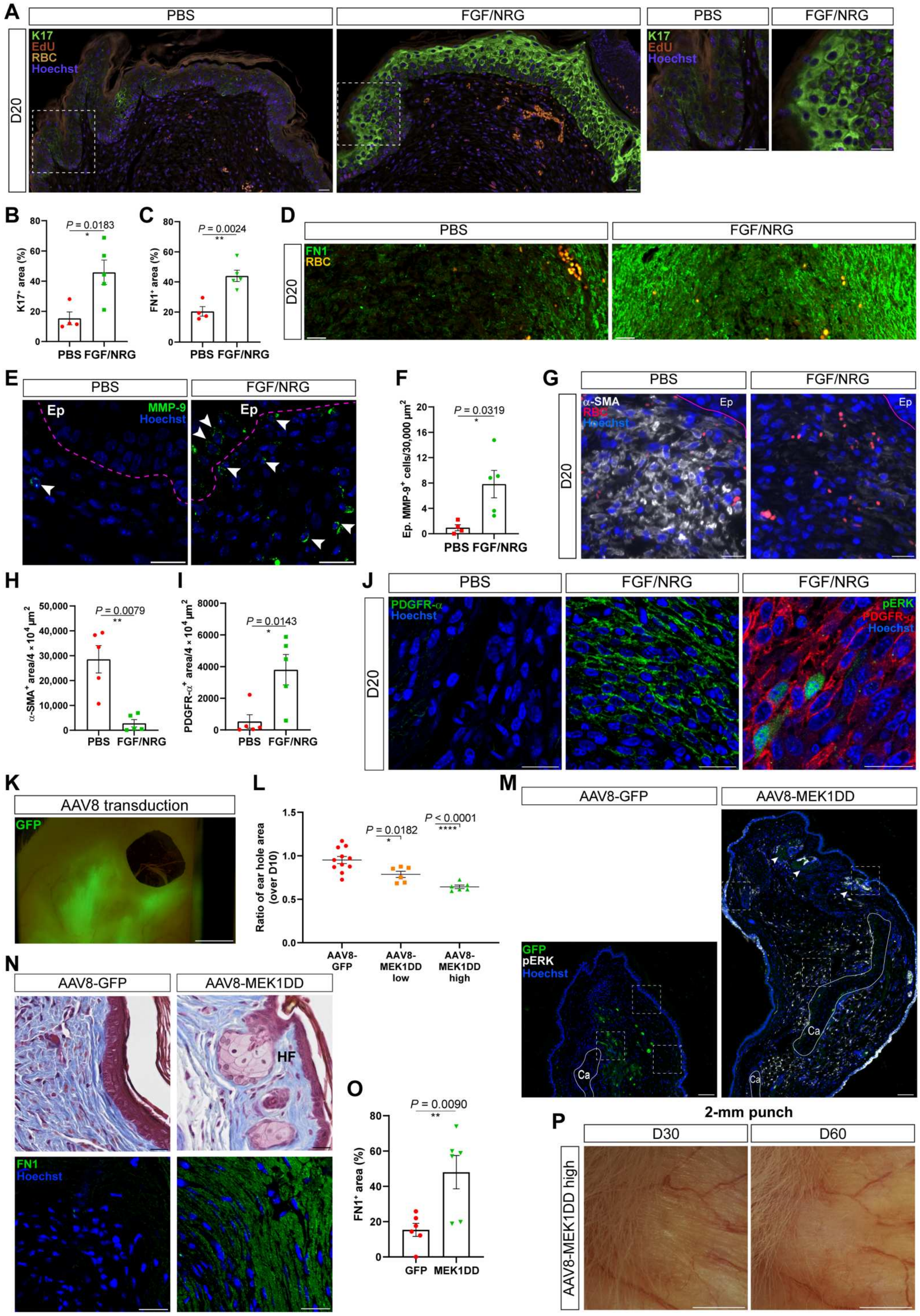

Fig. 10. Local ERK activation shifts fibrotic repair in *Mus* toward regenerative healing. (**A** to **J**) Data from bead implantation experiments. FGF-2/NRG-1– or PBS control–soaked microbeads were implanted at D10 (after re-epithelialization), and the tissue was collected at D20. Representative images and quantification of epidermal K17 (A and B) (green), zoom-in panels (A, right), FN1 (C and D), and MMP-9 (E and F) in the injury tip at D20 (n = 4 PBS, n = 5 FGF-2/NRG-1). Representative images and quantification of α-SMA$^+$ (G and H) and PDGFR-α$^+$ cells (I and J); co-labeling of pERK$^+$ and PDGFR-α$^+$ cells in FGF-2/NRG-1 group (J, right). (**K**) AAV8-GFP transduction of *Mus* ears showing GFP expression. (**L**) Quantification of *Mus*-transduced ear holes with AAV-GFP (control) and dose-dependent AAV-MEK1DD ($n \geq 6$ per group). (**M**) Representative images of ERK activation at the injury tip of AAV-transduced ears. (**N**) Representative images of Masson's trichrome (top) and FN1 (bottom) staining, showing newly formed hair follicles and enhanced FN1 deposition, respectively, upon AAV-MEK1DD induction (right) compared to GFP control, at the injury tip (positional information of the insets in fig. S20I). (**O**) Quantification of FN1-covering area at the injury tip of AAV-MEK1DD and AAV-GFP *Mus* wounds. (**P**) Representative images of ear hole closure after 2-mm punch in AAV8-MEK1DD–transduced *Mus* wounds. Epidermis (Ep)–dermis boundaries, pink line; autofluorescent red blood cells (RBC, orange); cartilage (Ca), hair follicle (HF),nuclei (Hoechst, blue). Scale bar, 20 μm, except 100 μm (M) and 2 mm (K and P). Each dot represents one biological replicate. Data are represented as means ± SEM; *$P < 0.05$, **$P < 0.01$, ****$P < 0.0001$.

transduction led to increased fibronectin deposition at the injury tip [Fig. 10, N (bottom), and O], which has been associated with a pro-regenerative milieu (*4*).

Next, we sought to investigate whether sustained ERK activity could induce full hole closure after a 2-mm ear punch. The 2-mm punch has been previously used to dissect some tissue regenerative properties (*52*, *74*). After transducing the AAV8-MEK1DD^{high} vector in the punched ears every 10 days, 2 of 6 mice closed the hole completely by D30 (Fig. 10P) and, overall, the hole diameter of 5 of 6 (83%) MEK1DD-transduced mice was ≤0.25 mm, while only 1 of 12 (8%) GFP-transduced control mice had a hole diameter of ≤0.25 mm at D60, indicating that sustained ERK activity can promote full regeneration of a 2-mm ear punch.

DISCUSSION

The mechanistic basis for naturally occurring examples of complex tissue regeneration in adult mammals remains incompletely understood. Here, we uncovered that ERK acts as a master regulator and molecular switch between tissue regeneration and fibrosis in the highly regenerative spiny mouse, a rare example of complex tissue regeneration in an adult mammal. Our data reveal that ERK activation is required for the early wound closure response and then specifically for tissue regeneration, where it (i) promotes a sustained proliferative response in stromal cells and keratinocytes and controls the expression of cyclins and E2F factors, (ii) induces an activated keratinocyte phenotype associated with a pro-regenerative wound epidermis, (iii) stimulates the expression of pro-regenerative ECM components, (iv) induces LEF1-dependent hair follicle neogenesis, and (v) transduces activated FGFR and ErbB-2 signaling within defined cell types (fig. S21).

Although ERK was immediately activated after injury in *Mus* and *Acomys* injuries, ERK activity was significantly more abundant in *Acomys* cells compared to *Mus*. This is consistent with the observation that *Acomys* fibroblasts have a more euchromatic state compared to the DNA of *Mus* cells (*4*), possibly resulting in a faster adaptation to extracellular stressors. Previous studies have shown that axolotls retain juvenile traits during adulthood by maintaining high levels of euchromatin, which keeps chromatin accessible for the prompt induction of gene expression in response to injury (*75*). Similar to quiescent stem cells, a yet-to-be-defined subset of *Acomys* cells appears to be retained in an alert state (*76*), allowing them to progress toward regenerative healing in response to injury cues. Consistent with this hypothesis, ERK-2 has been shown to bind developmental genes in embryonic stem cells and phosphorylate RNA polymerase II, which induces a poised state on gene promoters (*77*). Moreover, ERK phosphorylation can lead to chromatin accessibility through histone modifications (*43*) and the transcription of IEGs that are highly responsive to injury signals.

Our analysis of ERK-driven expression pattern during regeneration suggests that sustained high ERK activity after re-epithelialization is required to antagonize fibrosis by maintaining the expression of cell type–specific injury-responsive genes and activating regeneration-specific genes to sustain cell cycle progression and blastema formation. We found that the IEGs EGR-1 and Fos were induced in response to injury and their expression was sustained during *Acomys* regeneration. The number of EGR-1^+ and Fos^+ cells markedly decreased upon ERK, FGFR, and ErbB-2 inhibition. Hence, ERK acts as a control node where extracellular signals converge. The extended duration of ERK activity activates regeneration-specific effectors (i.e., Tnc, K5, and MMP-9) and sustains wound signals (i.e., K17, EGR-1, and Fos) to orchestrate a regeneration program at a cellular level. In addition, the ectopic application of ERK activators FGF-2/NRG-1 or AAV8-MEK1DD gene transfer was sufficient to stimulate *Acomys* regenerative traits in *Mus* injuries, including cell cycle reentry, pro-regenerative ECM remodeling, activated wound epidermis, and the ability to induce de novo hair follicle formation.

The analysis of ERK activity at a single-cell level revealed discrete cell subsets among the cell types populating the *Acomys* blastema, which are highly responsive to ERK activation. These include subpopulations of basal keratinocytes, fibroblasts, chondrocytes, endothelial cells, and immune cells, which may promote regeneration through cell proliferation (ERK-responsive cycling keratinocytes) and the secretion of pro-regenerative ECM molecules (ERK-responsive pro-regenerative fibroblast subset). Given the known function of ERK to induce chromatin accessibility and activate developmental genes (*43*, *77*), our findings show an increased expression of embryonic and stem cell markers (i.e. HMGA-2 and Snai1) in fibroblasts after FGF-2/NRG-1 treatment in *Mus* wounds, suggesting that ERK activity has the capacity to rewire mesenchymal cells toward a latent genetic regenerative program.

While many components of the ERK signaling pathway are evolutionarily conserved, the degree to which instances of tissue regeneration represent phylogenetic inertia or de novo episodes of convergent evolution remains unknown. Recent findings have revisited the hypothesis that there exists a "regeneration program," which is suppressed by inhibitory mechanisms promoting cell commitment and differentiation to safeguard tissue stability and DNA integrity (*66*, *78*). Natural selection may have favored mechanisms for fast wound closure and a strong fibrotic response to protect functional parenchyma at the expense of the plasticity required for regeneration. Thus, the ability to regenerate tissues and organs may not be completely lost but dormant in the cells of poorly regenerating mammals, such as adult laboratory mice, rats, and humans. Pro-regenerative components of MAPK/ERK signaling are activated in *Mus* cells, but their activity appears to be short-lived, thereby tapering off after wound closure. This supports the idea that fibrotic repair is present because a regenerative response is not sustained or secondarily initiated. However, extrinsic signals have the potential to trigger latent cell potential, circumvent regeneration blocks, and stimulate pro-regenerative traits. The idea that mammals may harbor dormant regenerative potential has obvious implications for regenerative medicine. The identification of specific factors that can activate regenerative signaling pathways holds the potential to awaken a latent regenerative phenotype of mammalian cells in damaged tissues and organs.

Our data establish the foundation for follow-up studies designed to identify the full complement of signaling inputs that can augment regenerative healing. Negative regulators of ERK activity, such as the DUSP phosphatases up-regulated in *Mus* injuries, could be ideal candidates to unleash ERK activation and promote regeneration. Similarly, future studies using our AAV8-MEK1DD or FGF-2/NRG-1 models in combination with lineage tracing would allow for a clearer understanding of whether increased proliferation and tissue growth has a clonogenic origin and address whether cell subsets reenter the cell cycle and exhaust their proliferative potential after a few cell divisions. These types of studies can inform whether subsequent molecular blocks impede further tissue growth or additional patterning cues are necessary to accomplish the regeneration of larger complex injuries. Ultimately, our data put forth a model in which ERK signaling acts as a major gatekeeper regulating fibrotic repair and regeneration in adult mammals.

MATERIALS AND METHODS

Animals

Spiny mice (*Acomys cahirinus*) and CD1 mice (*Mus musculus*) were housed at the University of Kentucky, Lexington, USA and at the Hubrecht Institute, Utrecht, the Netherlands, as previously described (*4*). At the Hubrecht Institute, spiny mice were kept in groups in individually ventilated cages (DoubleDecker System by Tecniplast). *Acomys* were fed a 3:1 mixture by volume of 14%

protein mouse chow and black-oil sunflower seeds. CD1 mice were fed mouse chow only. *Acomys* and *Mus* experienced a 12:12 (light: dark) light cycle. All the animals used were sexually mature and age- and sex-matched within experiments. All animal procedures were approved by the University of Kentucky Institutional Animal Care and Use Committee under protocol 2019–3254 and by the Animal Ethics Committee of the Royal Netherlands Academy of Arts and Sciences (KNAW) (project license AVD8010020187144).

Surgery

The 4-mm ear punch assay

Upon 4% (v/v) vaporized isoflurane-induced anesthesia, a through-and-through hole was made with a 4-mm biopsy punch in the center of the left and right ear pinna except for the proximal versus distal ear punch comparison described below. At specified time points, the animals were anesthetized and the ear hole area was calculated with a digital caliper, as previously described (*4*). Briefly, orthogonal diameters along the *y* and *x* axes were measured, the respective radii, r_y and r_x, were calculated, and the ear hole area (A) was obtained: $A = \pi r_y r_x$.

To detect differences in the ear hole closure rate between a proximal and a distal 4-mm ear punch, the proximal ear punch was performed at approximately 1 mm from the skull of the left ear and the distal ear punch at approximately 4 mm from the tip of the right ear along the *y* axis, both medially in the ear pinna of the same spiny mouse, as an internal control. The 4-mm punch assay was used in every experiment on the ear pinna except for Fig. 10P, where a 2-mm punch was used to investigate the full hole closure after MEK1DD transduction. Two-millimeter ear punches were made with 2-mm biopsy punches and placed in the center of the ear pinna.

The 8-mm back skin punch assay

Animals were anesthetized with isoflurane by inhalation. Anesthetized animals were shaved on the back. Next, metamizole (13 mg/kg body weight) was injected subcutaneously for pain relief. Two back skin punches per animal were made using an 8-mm-diameter biopsy punch, resulting in full-thickness excisional wounds. The animals were allowed to recover on a warm plate and monitored for the following days. The animals were euthanized at specific time points when the tissue was collected.

Myocardial infarction model

Infarct of the myocardium was induced in adult spiny mice through permanent ligation of the left anterior descending artery, as previously described (*16*).

In vivo treatments

After 4-mm ear punch, animals were randomized and subjected to intraperitoneal injections of a pERK inhibitor, FGFR inhibitor, ErbB-2 inhibitor, or vehicle. Pharmacological ERK inhibition was achieved by injecting the highly selective MEK inhibitor mirdametinib (PD; purchased by Selleck Chemicals). In all the experiments (Early, Continuous, and Late PD), PD was injected at 18 mg/kg in 200 μl of total volume per mouse every other day. For the experiments at D15 (the RNA-seq, RNA-seq data validation, scRNA-seq, and immunostainings), PD was injected daily from D12 to D15 at the same dose and formulation. The control group was injected with the same mix, except for the inhibitor. The Continuous PD group was treated from day −1 across the duration of the experiment. The Early PD group was treated from day −1 to D20. The Late PD group was treated from D12 until the end of the experiment. The selective AZD4547 (7 mg/kg) (*50*) and mubritinib (TAK165, 10 mg/kg) (*51*) small molecules were used to pharmacologically inhibit the FGFR and ErbB-2 receptors, respectively. They were injected as the pERK inhibitor in 200 μl of total volume per mouse daily from D12, after re-epithelialization had occurred, to D20. For the scRNA-seq experiment, the pERK, FGFR, and ErbB-2 inhibitors were administered from D12 to D15 only. All the inhibitors were dissolved in a mix of dimethyl sulfoxide, polyethylene glycol, and PBS/Tween upon injection based on the manufacturer's recommendations.

For the rescue studies in *Mus*, after 4-mm ear punches, the animals were randomized and subjected to implantation of microbeads (Affi-Gel Blue beads, Bio-Rad, approximately 150 μm diameter) soaked with ERK activators: FGF-2 (Sigma-Aldrich, no. 341583)/NRG-1 (PeproTech, no. 100-03) (1.25 and 2 μg/μl, respectively, dissolved in PBS and evenly mixed). PBS-soaked beads were used as controls. Two soaked microbeads were implanted in the ear punch area at D10 after re-epithelialization occurred, and the ear tissue was collected at D13 or D20 for immunohistochemistry. In the same way, two microbeads were implanted at D10 and D15, and the tissue was collected at D20 for the assessment of ear hole closure and hair follicle neogenesis. The respective PBS control group underwent the same procedure.

To assess the number of cells in S phase in *Acomys* and *Mus* wounds, 2 mM EdU diluted in 200 μl of saline solution was intraperitoneally injected through three EdU pulses evenly distributed over a period of 9 hours before tissue collection. EdU-incorporating cells were detected on formalin-fixed, paraffin-embedded tissue sections through copper-catalyzed click reaction by using Alexa Fluor 594 Azide (Thermo Fisher Scientific).

For the AAV8-mediated gene transfer experiments, 10 days after the 4-mm ear punch, AAV8-MEK1DD or the AAV8-GFP viral particles were injected locally at the base of the ear punch through a Hamilton syringe while animals were under anesthesia. The low-dose group received approximately 10^9 gc, and the high-dose group received 5×10^{10} to 5×10^{11} gc ($n \geq 6$ per group). In the hole closure experiment after 2-mm punch, the AAV8-MEK1DD^{high} and AAV8-GFP control vectors were injected in the same way as described above, every 10 days until the experimental end point.

Histology and immunohistochemistry

Following tissue collection, samples were fixed overnight in 10% (v/v) neutral-buffered formalin at 4°C and washed in PBS, followed by dehydration and paraffin embedding. Five-micrometer-thick ear sections were cut and hydrated. Sections were subjected to Masson's trichrome staining according to the manufacturer's instructions or immunohistochemistry by using antibodies listed in table S3. Heat-mediated antigen retrieval with high pH buffer solution was performed before staining. Normal goat serum (10%, Vector Laboratories, S-1000) was used as blocking buffer, except for the goat anti–PDGFR-α antibody for which bovine serum albumin (BSA; 5%) was used. Primary antibodies were visualized with Alexa Fluor 488, 594, or 647 secondary antibodies or with Alexa Fluor 488– or Alexa Fluor 647–conjugated streptavidin antibodies (Invitrogen, 1:1000) after incubation with biotinylated secondary antibodies (Vector Laboratories, 1:500). Streptavidin/Biotin blocking kit (SP-2002, Vector Laboratories) was used in case of fluorophore-conjugated streptavidin detection. Nuclei were counterstained with Hoechst (10 mg/ml, diluted to 1:7500). ProLong Gold Antifade (Invitrogen) was used as mounting medium. Negative controls with either no primary antibody or corresponding immunoglobulin G (IgG) isotypes were used. Back skin samples were fixed for 2 hours in 4% paraformaldehyde at room temperature, washed in PBS, and processed as the ear samples. Seven-micrometer-thick back skin sections were cut and hydrated. Heart samples were collected and processed as previously described (*16*).

Bright-field images and Masson's trichrome staining were acquired on a Leica DM4000 microscope and Olympus VS200 slide scanner. Immunofluorescent images were acquired with Leica Thunder Imager, Zeiss LSM700, or LSM900 confocal microscopes. Experimental samples were acquired and processed in the same way. Cell counting, area, and signal quantification were performed on multiple regions of interest (ROIs) per sample using ZEN (version 2.6 lite) and ImageJ software (version 1.52p). Several photomicrographs of the injury area were obtained at 20×, 40×, or 63× magnification using LSM700, LSM900, or Leica Thunder Imager. Signal quantification was performed on the experimental samples by subtracting the respective background signal from each tissue section and thresholding the fluorescent signal with ImageJ software (version 1.52p). The area is expressed as percentage of the area of positive signal over the total area of interest. The quantification from the different micrographs within the sections was averaged to give the final value per sample. When quantifying epidermal markers, only the epidermal area was considered. Conversely, when quantifying the positive signal in the dermis, only the dermal compartment of the injury area was used. The quantification of cell number is expressed as percentage of positive nuclei over the total number of nuclei per area of interest. The number of LEF-1$^+$ placodes indicates the average of LEF-1$^+$ cell clusters localized in the epidermis. To assess the pERK antibody and rule out any interspecies variability of antibody binding, the pERK max pixel value/ROI (10^4 μm^2) was quantified; each dot corresponds to an ROI with its relative pERK max value (0 to 255 pixel values). Although *Acomys* pERK signal is overall higher when compared to *Mus*, *Mus* samples also have ROIs with pERK max values similar to the *Acomys* counterparts (i.e., *Mus* at 6 hpi) (fig. S1B), ruling out a potential technical difference in the pERK antibody efficiency across species. The quantification of total ERK activation over time between *Mus* and *Acomys* was normalized to pERK levels at day 0 (uninjured) within each species.

Western blot
Ear tissue was collected with an 8-mm biopsy punch circumscribing the original 4-mm ear punch at specified time points and lysed in radioimmunoprecipitation assay buffer containing protease inhibitors. Total proteins were quantified using a bicinchoninic acid assay (Thermo Fisher Scientific), and equal amounts of proteins were run on a gradient polyacrylamide gel (Mini-PROTEAN precast gels, Bio-Rad). Proteins were transferred to polyvinylidene difluoride membranes, and membranes were blocked in BSA (5%) and probed with the following antibodies: pERK (Cell Signaling Technology, no. 4370, 1:2000) and β-tubulin (Sigma-Aldrich, T4026, 1:500). Primary antibodies were detected with horseradish peroxidase–conjugated secondary antibodies and enhanced chemiluminescence (Thermo Fisher Scientific).

Bulk tissue transcriptomics
Half-rings of tissue from the proximal punched ear segments were collected from sexually mature and age- and sex-matched *Acomys* for gene expression profiling at 24 hpi and D15 after pharmacological ERK inhibition. The tissue was snap-frozen, and total RNA was extracted in TRIzol (Life Technologies) according to the manufacturer's instructions and processed for QuantSeq 3′mRNA-seq from Lexogen Services (https://lexogen.com/services). Briefly, alignment was performed with the Spliced Transcripts Alignment to a Reference (STAR) tool. The RNA-seq Quality Control package (RseQC) was run to comprehensively evaluate the RNA-seq data. RNA integrity and concentration were measured by Bioanalyzer Systems (from Agilent Technologies) and Qubit Fluorometer, respectively. A total of 17 samples were sequenced with the following distribution: $n = 4$/control 24 hpi, $n = 3$/PD 24 hpi, $n = 5$/control D15, $n = 5$/PD D15. Read counts were processed with iDEP.92 (integrated Differential Expression and Pathway analysis), and DEGs were identified using DESeq2 (*79*) for comparisons at each time point (FDR cutoff, 0.10).

The volcano plot (Fig. 3B) shows statistical significance ($-\log_{10}$ adjusted *P* value) plotted against $\log_2$ fold change expression values. Genes were down-regulated (negative values) and up-regulated (positive values) in PD. Color-coded genes were categorized as follows: cell cycle (green), activated epidermis (orange), matrisome (red), immune response (blue), and nerve growth (pink).

GSEA was performed with GSEA software (v4.1.0) and the Molecular Signaling Database (MSigDB, v7.2) (*54*, *80*). The upper part of the GSEA plot shows the enrichment score (ES), and the normalized ES (NES) (expression dataset versus ranked list of genes) is indicated together with the FDR *q* and nominal *P* value. The *x* axis indicates the genes (vertical black lines) represented in gene sets. The bottom part represents degree of correlation for genes with the two represented groups: positive correlation (red) and negative correlation (blue).

The heatmap of the ERK-dependent matrisome was generated by interrogating the murine ECM atlas (http://matrisomeproject.mit.edu/ecm-atlas/) with the top DEGs (FDR < 0.1) of the PD versus control dataset at D15. The matrisome is divided into matrisome-associated genes and core matrisome genes. The matrisome-associated division was subcategorized into secreted factors (pink), ECM-affiliated proteins (dark orange), and ECM regulators (light orange). The core matrisome included proteoglycans (cyan), ECM glycoproteins (purple), and collagens (cobalt blue).

The ERK-dependent AP-1 target genes were identified by interrogating the Transcription Factor Target (TFT) collection (MSigDB) and TF binding site database by QIAGEN with the top DEGs between control and PD-treated *Acomys*. The identified genes contain at least one AP-1 binding site (TGANTCA) in their promoter region. The genes encoding transcription factors were identified through the Gene families analysis of the MSigDB gene sets.

To validate the RNA-seq data, RNA was reverse-transcribed using SuperScript IV VILO Master Mix with ezDNAse Enzyme (Thermo Fisher Scientific) and SYBR Green quantitative polymerase chain reaction (PCR) was performed on a Roche LightCycler multi-well plate 384 with species-specific primers enlisted in table S4, designed using the National Institutes of Health (NIH) Primer-Blast tool (*81*). Spiny mouse transcripts were accessed through the SequenceServer BLAST database (*82*). Gene expression was analyzed using ΔΔCt method and normalized against *Tbp* housekeeping gene (encoding TATA-box binding protein).

Single-cell transcriptomics

The punched ear tissues were obtained from *Acomys* and *Mus* wounds at D15. Only tissue proximal to the injury was collected with an 8-mm biopsy punch from both the left and right ears and by pooling the tissue from four age- and sex-matched animals. After collection, the tissue was placed in cold HypoThermosol FRS (Sigma-Aldrich, H4416) and minced thoroughly with a razor blade in a petri dish on ice. The minced tissue was placed in the *Bacillus licheniformis* enzyme mix (10 mg/ml protease, Sigma-Aldrich, P5380 and 0.5 M EDTA dissolved in Dulbecco's PBS) according to published studies (*83*). The tissue was digested on ice for 20 min and homogenized in a 2-ml KIMBLE Dounce tissue grinder (Sigma-Aldrich, D8938) for 8 min. The supernatant containing cells was filtered through a 30-μm MACS SmartStrainer (Miltenyi Biotec, 130-098-458). The remaining tissue was ground in a fresh enzyme mix for eight additional minutes, passed through a 30-μm strainer, and added to the previously filtered suspension. Cells were centrifuged at 300*g* for 5 min at 4°C and resuspended in ice-cold PBS/BSA 0.04%. Red blood cells (RBCs) were removed through the RBC lysis buffer Hybri-Max (Sigma-Aldrich, R7757). The suspension was centrifuged at 300*g* for 5 min at 4°C and resuspended in ice-cold PBS/BSA 0.04%. Cell count and viability was assessed using a hemocytometer with trypan blue.

The barcoding of individual transcripts from single cells, followed by cDNA amplification and library construction, were performed using 10x Genomics reagents and 10X Chromium Controller according to the manufacturer's user guide nr. CG000315 ("Chromium Next GEM Single Cell 3′ Reagent Kits v3.1 (Dual Index), RevC"). Amplified cDNA and final libraries were subjected to quality control on an Agilent 2100 Bioanalyzer using the High Sensitivity DNA Reagent Kit. Libraries were sequenced on NovaSeq 6000 instrument using NovaSeq 6000 S1 Reagent Kit v1.5 (Illumina).

The *A. cahirinus* assembly and gene annotation were generated from the INSDC Assembly GCA_907164435.1 using the mouse ortholog gene names when there was a one-to-one match. For *A. cahirinus*, the genome indexes were created with Cell Ranger version 6.1.1 (10x Genomics) using the "mkref" mode and default settings. For *M. musculus*, the refdata-gex-mm10-2020-A indices were obtained (10x Genomics). For mapping reads to the genome and tallying into count tables, Cell Ranger count was used with default settings. The Seurat package (version 4.0.2, PMID: 26000488) was used to identify cell clusters and perform differential gene expression based on UMAP. The algorithm used to cluster cells was implemented and performed using Seurat, with a cutoff of 200 expressed genes per cell in at least three cells. A total of 2000 features were returned per dataset and used for downstream analysis. The built-in features used to recognize significant principal components (PCs) were used and visually inspected. The first 14 PCs were selected for subsequent UMAP clustering. Cells with less than 500 detected genes or more than 45,000 (possible duplicates) were omitted for downstream analysis, as well as all cells with more than 2.5% hemoglobin-related genes. Mitochondrial filtering was not applied because of missing annotation in *Acomys*. When merging all datasets together, batch-corrected dimension reduction was performed by merging the datasets using 2000 anchor genes common to the entire dataset within the FindIntegrationAnchors function, followed by a final merging using the IntegrateData function. Dimensionality reduction was performed using UMAP. Cell clusters were determined using the Louvain algorithm within the Seurat FindClusters function. Marker genes were determined for each cluster based on the nonparametric Wilcoxon rank sum test within the Seurat FindMarkers function.

Statistics

The normality of residuals was tested to check the Gaussian data distribution, and the *F* test was used to assess the equality of variances. Differences across groups were evaluated for statistical significance using two-sided Student's *t* tests when comparing two groups normally distributed and with equal variances. Unpaired *t* test with Welch's correction was used when comparing two groups normally distributed with unequal variances. Spearman or Pearson bivariate analyses were used for correlation studies. A two-sided Mann-Whitney test was used when comparing two groups not following a normal distribution. Kruskal-Wallis test with Dunn's post hoc test was used when comparing more than two groups non-normally distributed. One-way ANOVA with Dunnett's post hoc test was used to compare more than two groups normally distributed. Two-way repeated-measures (RM) ANOVA with Sidak's post hoc test was used to compare more than two groups normally distributed over time or distance. Paired analyses were performed with a two-tailed paired *t* test. Potential outliers were defined by Grubbs' test ($\alpha = 0.05$). Graphs show means ± SEM. *P* values are shown in the graphs. Statistical significance was assessed as ns (not significant), $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$, or $^{****}P < 0.0001$. Sample size for each experiment is indicated in the figure legends. All statistical analyses were performed with GraphPad Prism software (v8.1.1).

Supplementary Materials

This PDF file includes:
Figs. S1 to S21
Tables S1 to S4

View/request a protocol for this paper from *Bio-protocol*.

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An ERK-dependent molecular switch antagonizes fibrosis and promotes regeneration in spiny mice (*Acomys*)

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